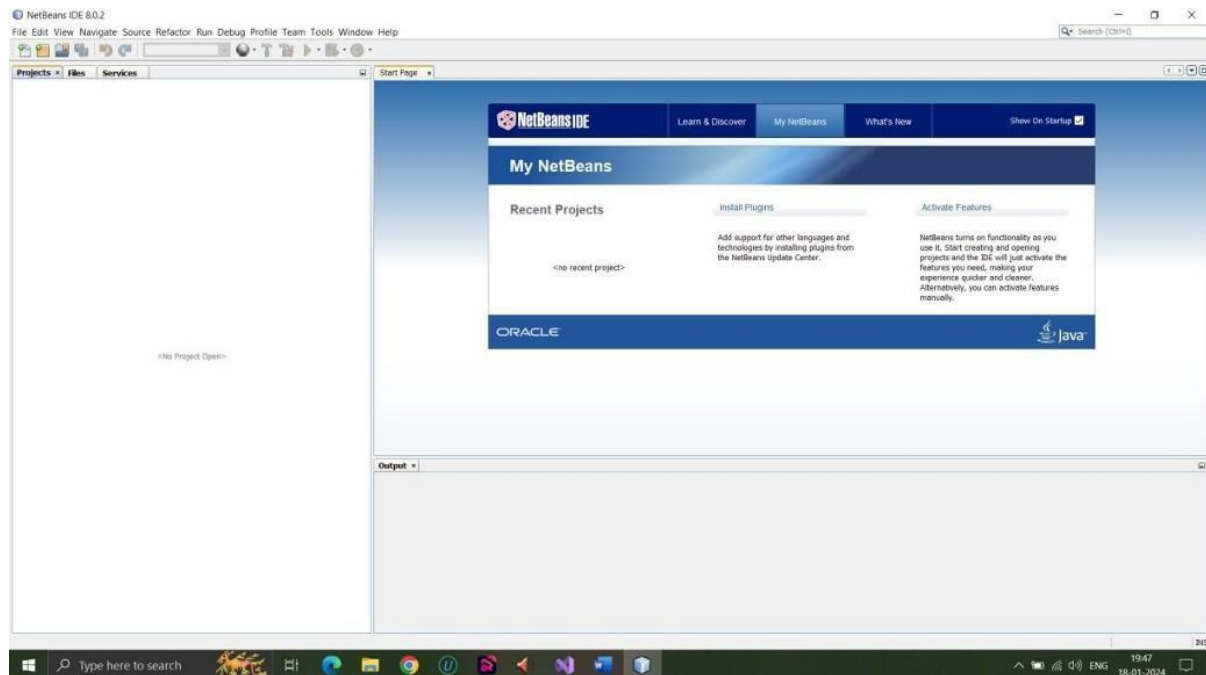


Practical 1

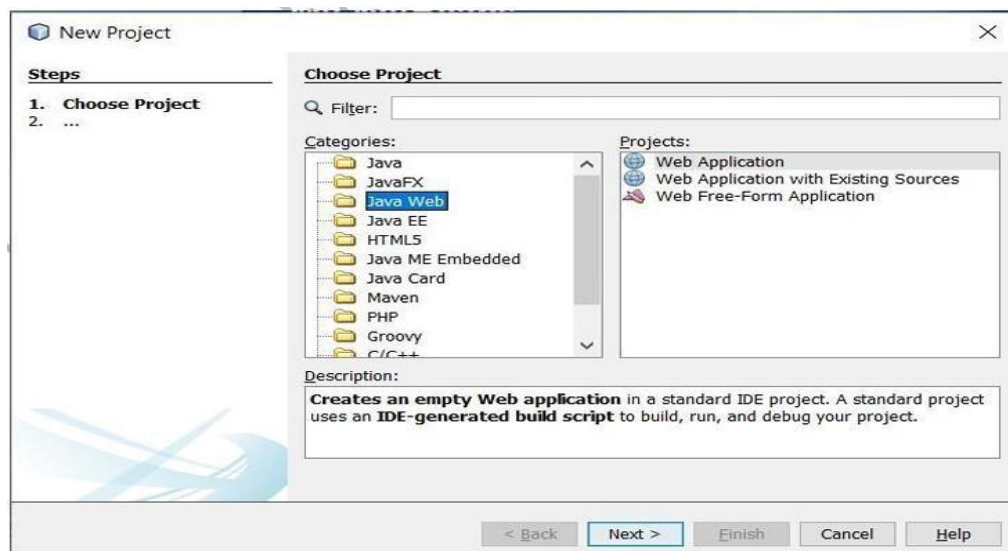
AIM: Define a simple services like Converting Rs into Dollar and Call it from different platform like JAVA and PYTHON.

Steps:

1] Open the NetBeans , and you will get the following screen. Close the startpage.



2] Nowclick on the file tab and click on new project you will get the following screen:



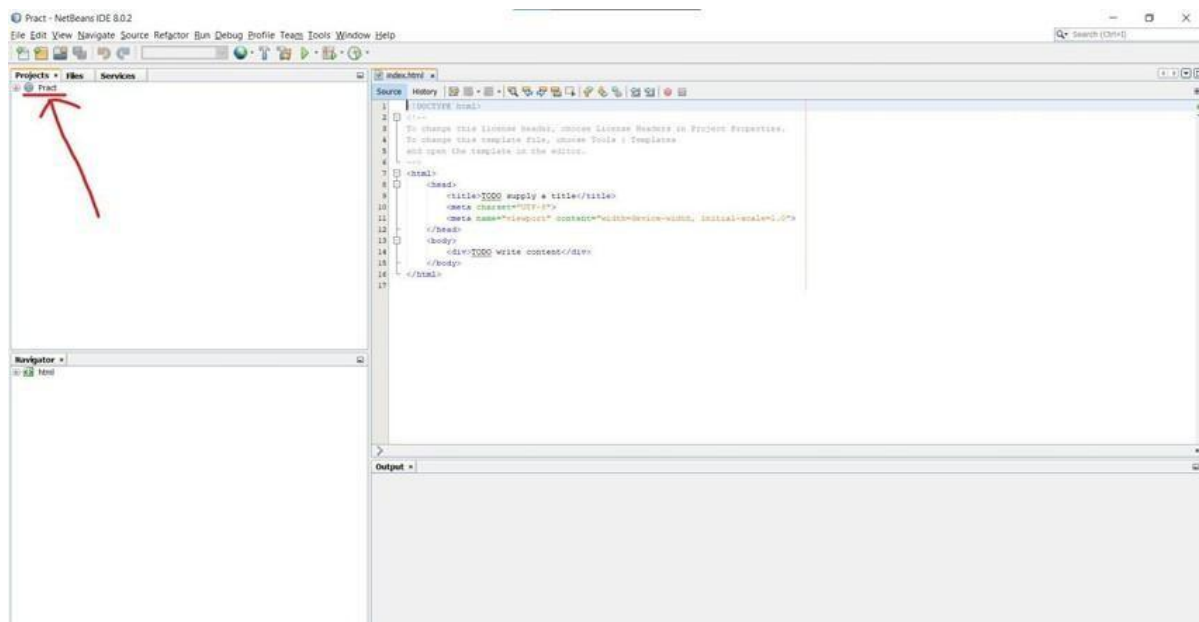
3] In Categories select Java Web and in Projects , Select Web Application .After selecting click onnext . You will get the following window:

The screenshot shows the 'New Web Application' dialog box with the 'Name and Location' tab selected. The 'Steps' list on the left indicates the current step is '2. Name and Location'. The 'Project Name' field is filled with 'Pract1'. The 'Project Location' field is filled with 'C:\Users\LENOVO\Documents\NetBeansProjects' and has a 'Browse...' button next to it. The 'Project Folder' field is filled with 'C:\Users\LENOVO\Documents\NetBeansProjects\Pract1'. There is an unchecked checkbox for 'Use Dedicated Folder for Storing Libraries' with a 'Libraries Folder' field and a 'Browse...' button next to it. A message states: 'Different users and projects can share the same compilation libraries (see Help for details)'. A red error message at the bottom says: 'Project Folder already exists and is not empty.' The bottom navigation bar contains buttons: '< Back', 'Next >', 'Finish', 'Cancel', and 'Help'.

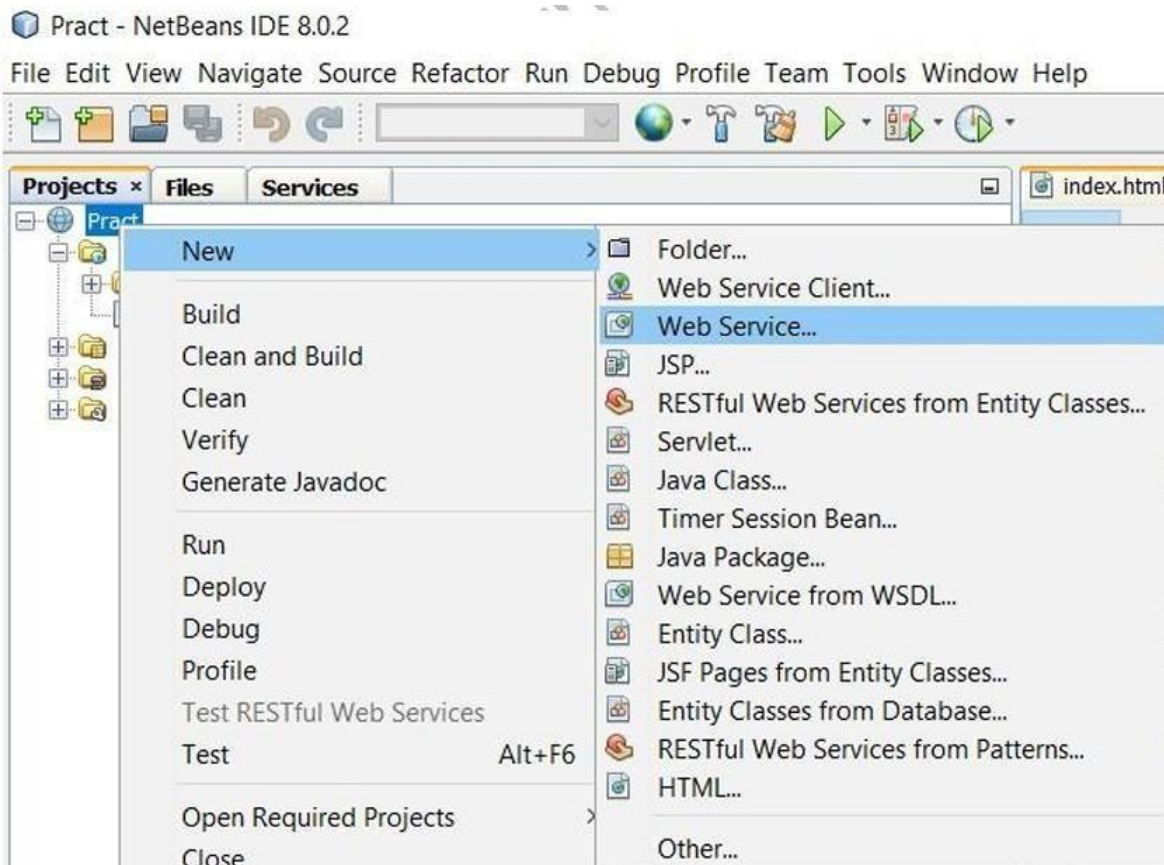
4] Now Give name to the Project Name: , and click on next . you will get the following window then again click on finish:

The screenshot shows the 'New Web Application' dialog box with the 'Server and Settings' tab selected. The 'Steps' list on the left indicates the current step is '3. Server and Settings'. The 'Add to Enterprise Application' dropdown is set to '<None>'. The 'Server' dropdown is set to 'GlassFish Server 4.1' and has an 'Add...' button next to it. The 'Java EE Version' dropdown is set to 'Java EE 7 Web'. The 'Context Path' field is filled with '/Pract'. The bottom navigation bar contains buttons: '< Back', 'Next >', 'Finish', 'Cancel', and 'Help'.

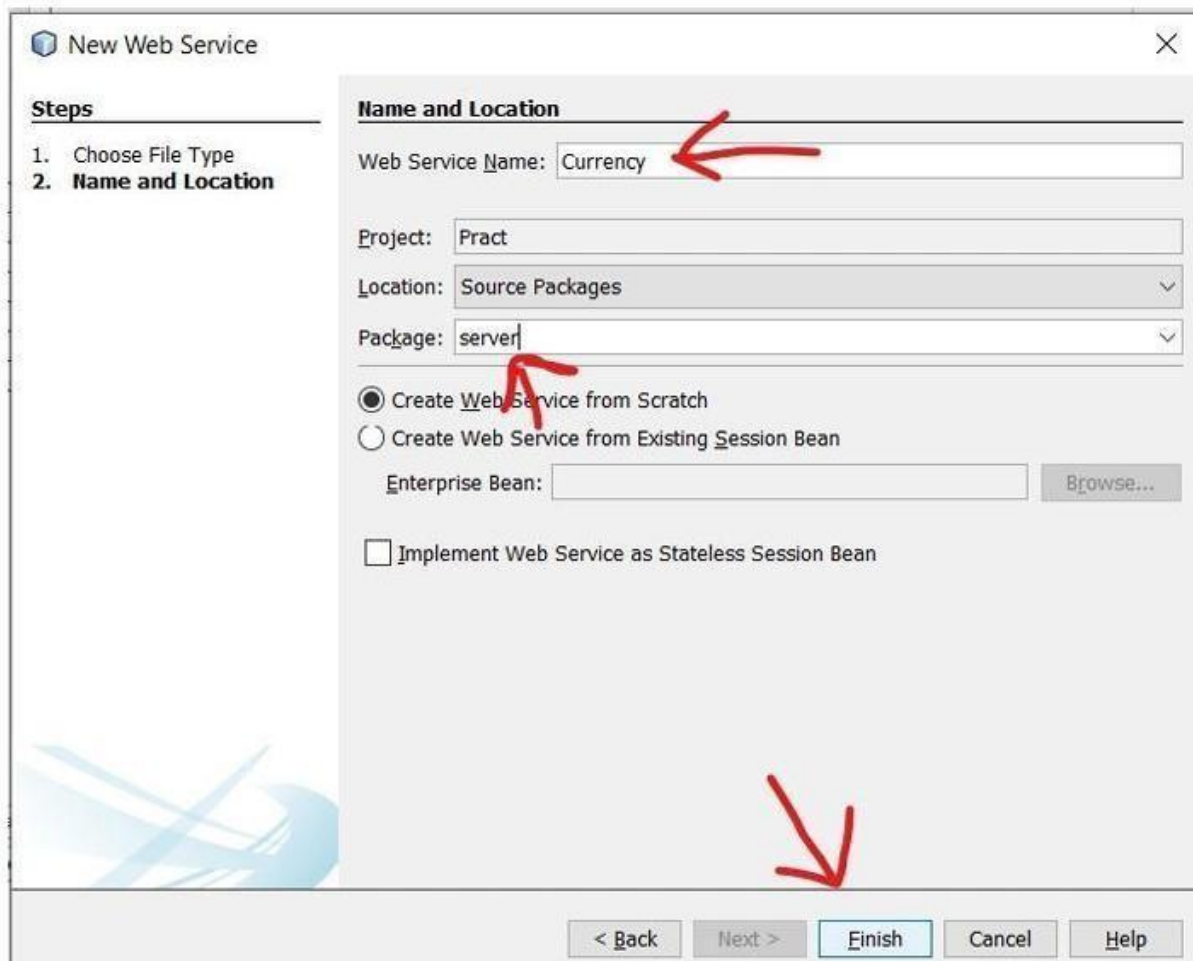
5] You will get the following screen now carefully see in projects section your recently created project appears double click to expand it:



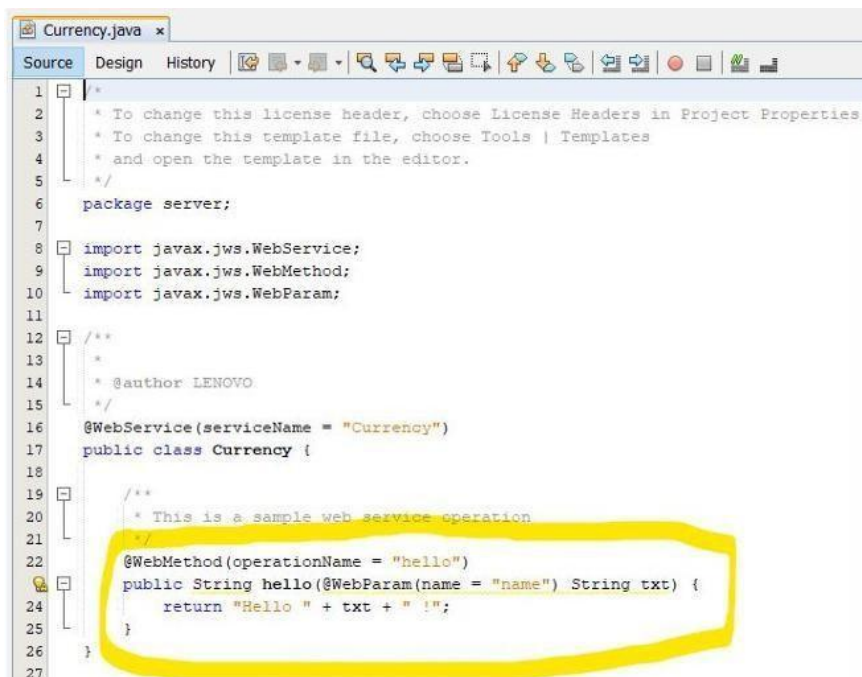
6] Now Right click on the project and select new and then select Web Service ,As shown Below:



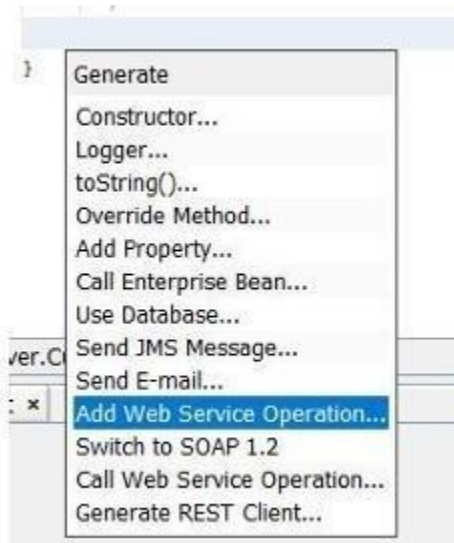
7] after clicking Web Service following window should appear , now give name to web service and give package as "server". As shown in the image:



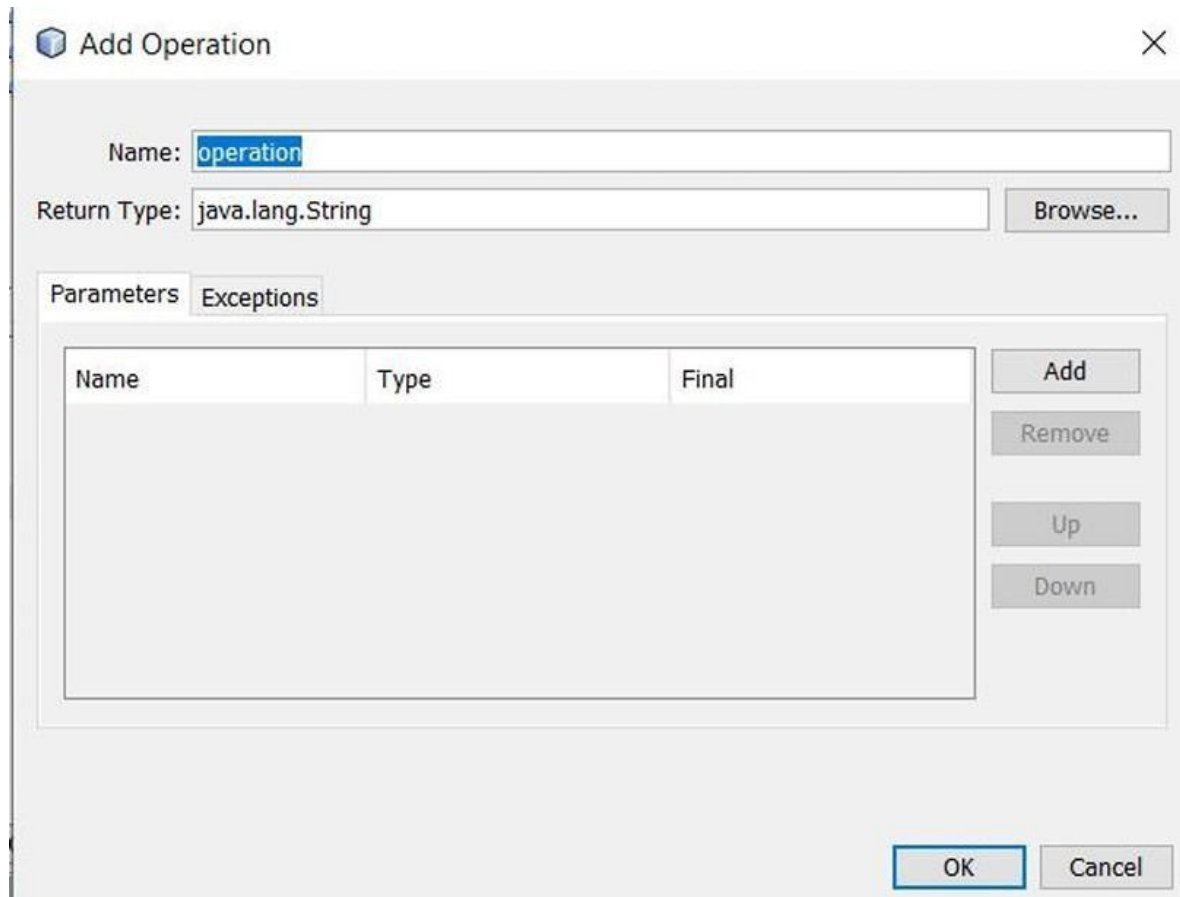
After clicking finish you should get the following window erase the mentioned code :



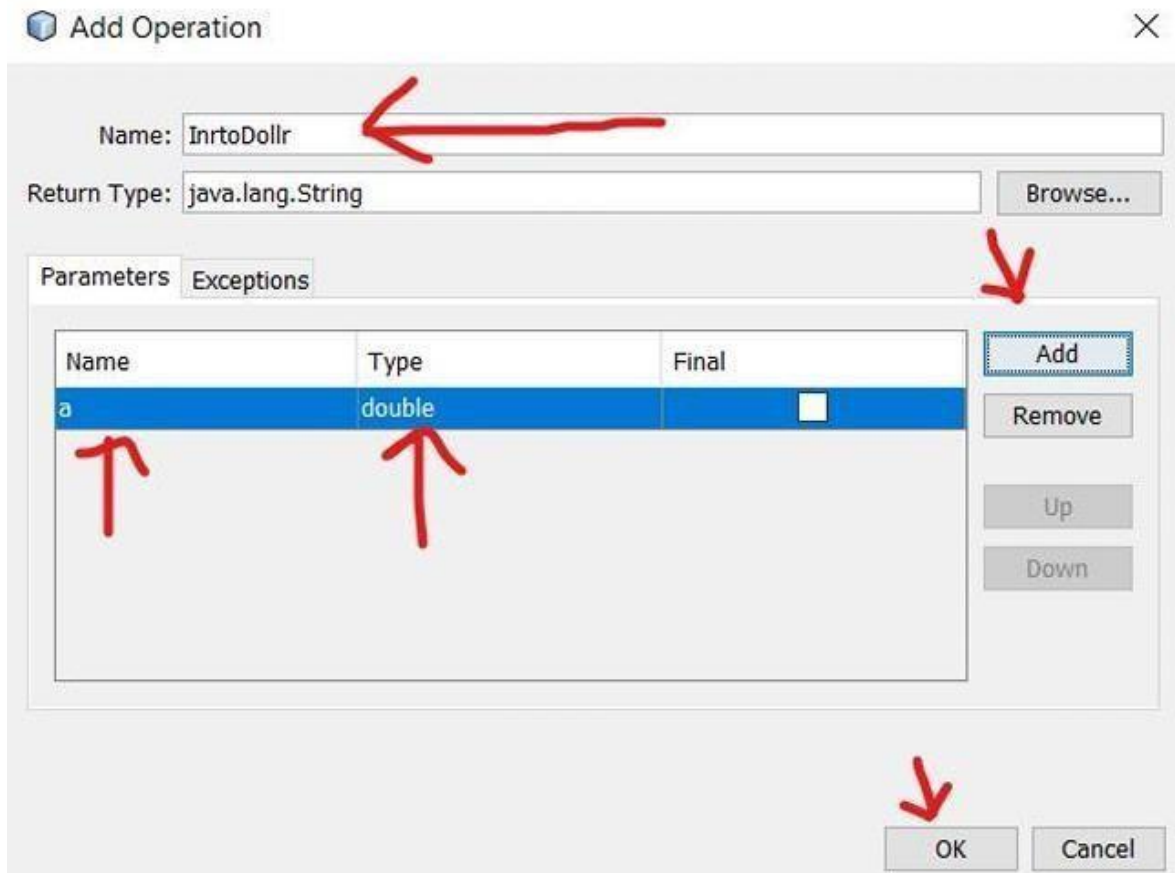
8] Nowright click any where and click on insert code and select Add Web Service Operation:



9] the following window would appear: Just give name to the method or operation and click on add button to add parameters to the method as here we are converting dollar to rupees we should need only one parameter .



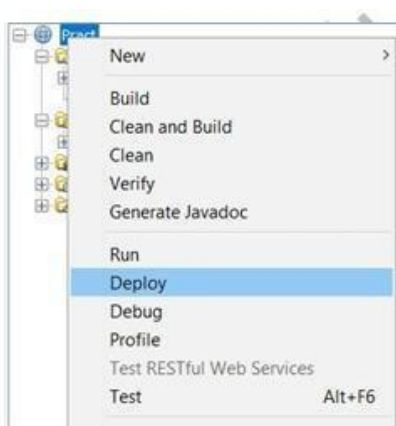
10] give the name to the variable select data type as double and click on ok as shown below:



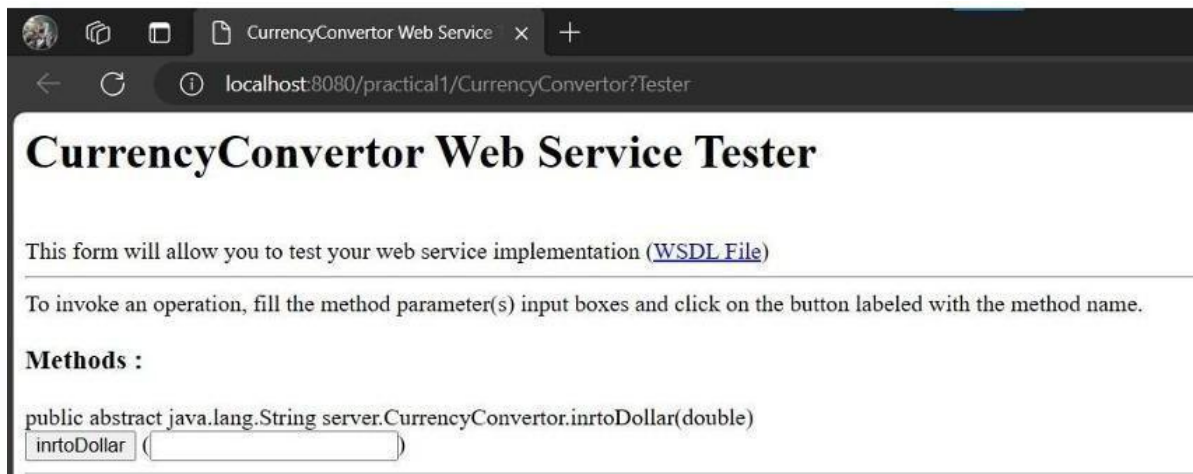
11] After clicking ok code will autogenerate , make changes In that code as mentioned below:

```
@WebMethod(operationName = "InrtoDollar")
public String InrtoDollar(@WebParam(name = "a") double a) {
    //TODO write your implementation code here:
    return "The Indian rupees "+a+" in Dollars is "+(a/83.17);
}
```

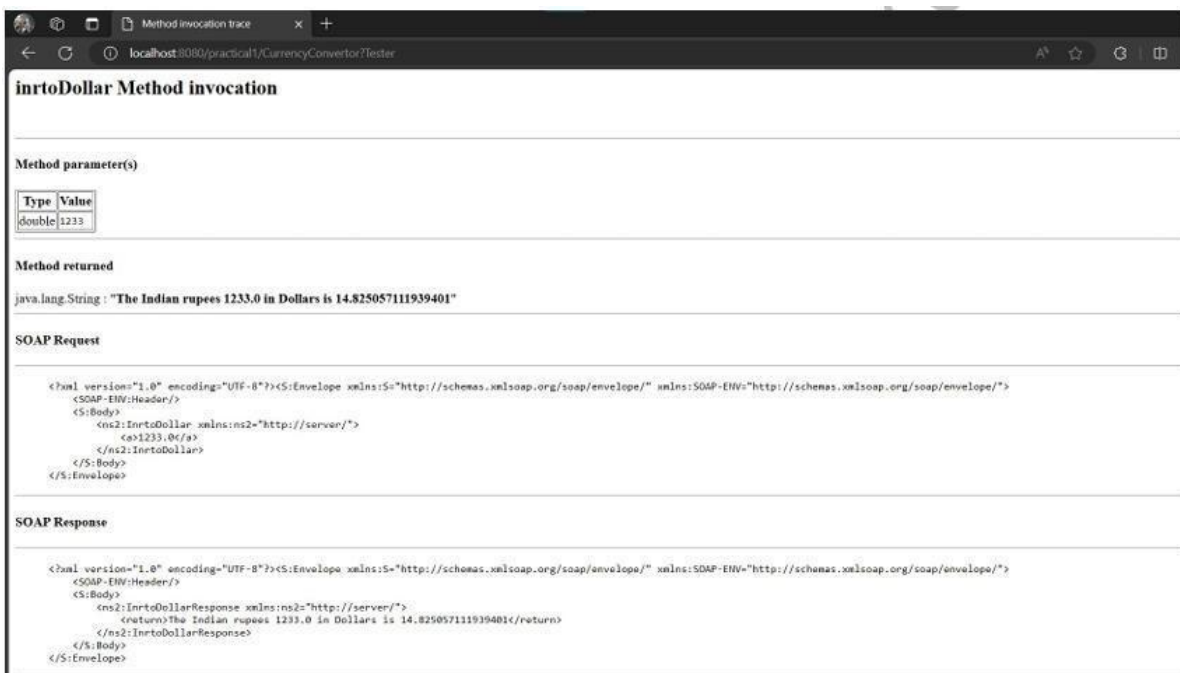
12] now our webservice is ready now right click on project and click on deploy:



13] now right click on webservice and click on test web service you will get the following output:



The screenshot shows a web browser window with the title "CurrencyConvertor Web Service". The address bar shows "localhost:8080/practical1/CurrencyConvertor?Tester". The main heading is "CurrencyConvertor Web Service Tester". Below the heading, there is a text area that says "This form will allow you to test your web service implementation ([WSDL File](#))". Below this, there is a text area that says "To invoke an operation, fill the method parameter(s) input boxes and click on the button labeled with the method name." Below this, there is a section titled "Methods :". Under "Methods :", there is a text area that says "public abstract java.lang.String server.CurrencyConvertor.inrtoDollar(double)". Below this, there is a text input field with the value "1233.0" and a button labeled "inrtoDollar".



The screenshot shows a web browser window with the title "Method invocation trace". The address bar shows "localhost:8080/practical1/CurrencyConvertor?Tester". The main heading is "inrtoDollar Method invocation". Below the heading, there is a section titled "Method parameter(s)". Under "Method parameter(s)", there is a table with two columns: "Type" and "Value". The table has one row with "double" in the "Type" column and "1233" in the "Value" column. Below the table, there is a section titled "Method returned". Under "Method returned", there is a text area that says "java.lang.String : 'The Indian rupees 1233.0 in Dollars is 14.825057111939401'". Below this, there is a section titled "SOAP Request". Under "SOAP Request", there is a text area that shows the SOAP request XML:

```
<?xml version='1.0' encoding='UTF-8'?><S:Envelope xmlns:S='http://schemas.xmlsoap.org/soap/envelope/' xmlns:SOAP-ENV='http://schemas.xmlsoap.org/soap/envelope/'>
  <SOAP-ENV:Header/>
  <S:Body>
    <ns2:InrtoDollar xmlns:ns2='http://server/'>
      <in>1233.0</in>
    </ns2:InrtoDollar>
  </S:Body>
</S:Envelope>
```

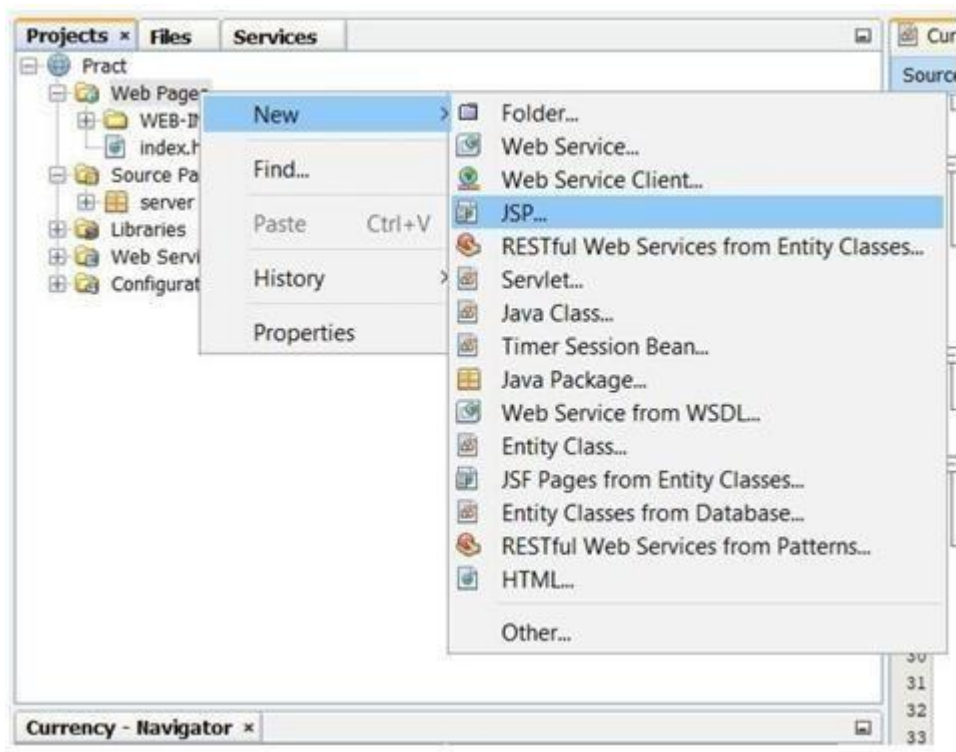
 Below this, there is a section titled "SOAP Response". Under "SOAP Response", there is a text area that shows the SOAP response XML:

```
<?xml version='1.0' encoding='UTF-8'?><S:Envelope xmlns:S='http://schemas.xmlsoap.org/soap/envelope/' xmlns:SOAP-ENV='http://schemas.xmlsoap.org/soap/envelope/'>
  <SOAP-ENV:Header/>
  <S:Body>
    <ns2:InrtoDollarResponse xmlns:ns2='http://server/'>
      <return>The Indian rupees 1233.0 in Dollars is 14.825057111939401</return>
    </ns2:InrtoDollarResponse>
  </S:Body>
</S:Envelope>
```

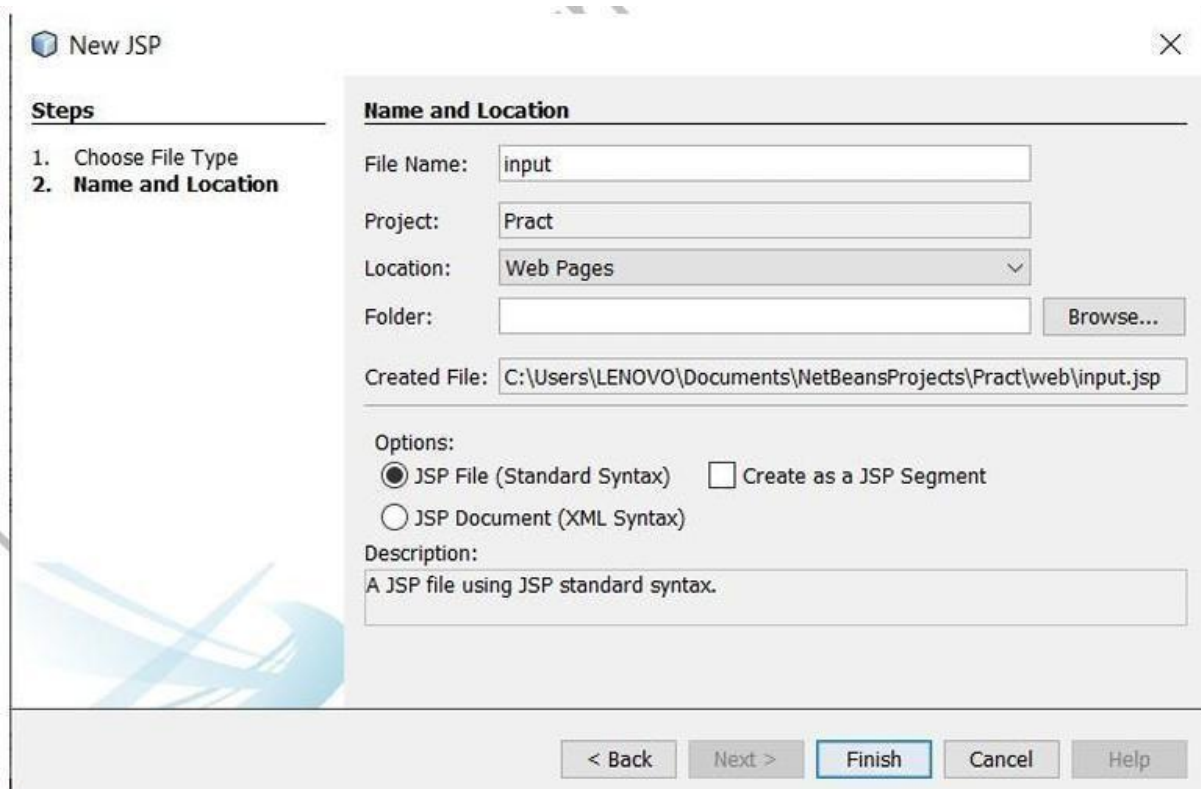
So this is how we created our web service and deployed it.

Creating a java Client using jsp

14] now our web service is successfully deployed. Right click on web pages and select new and select jsp as shown below:



15] give name and click on finish as shown below do it 2 times on for input and one for output:



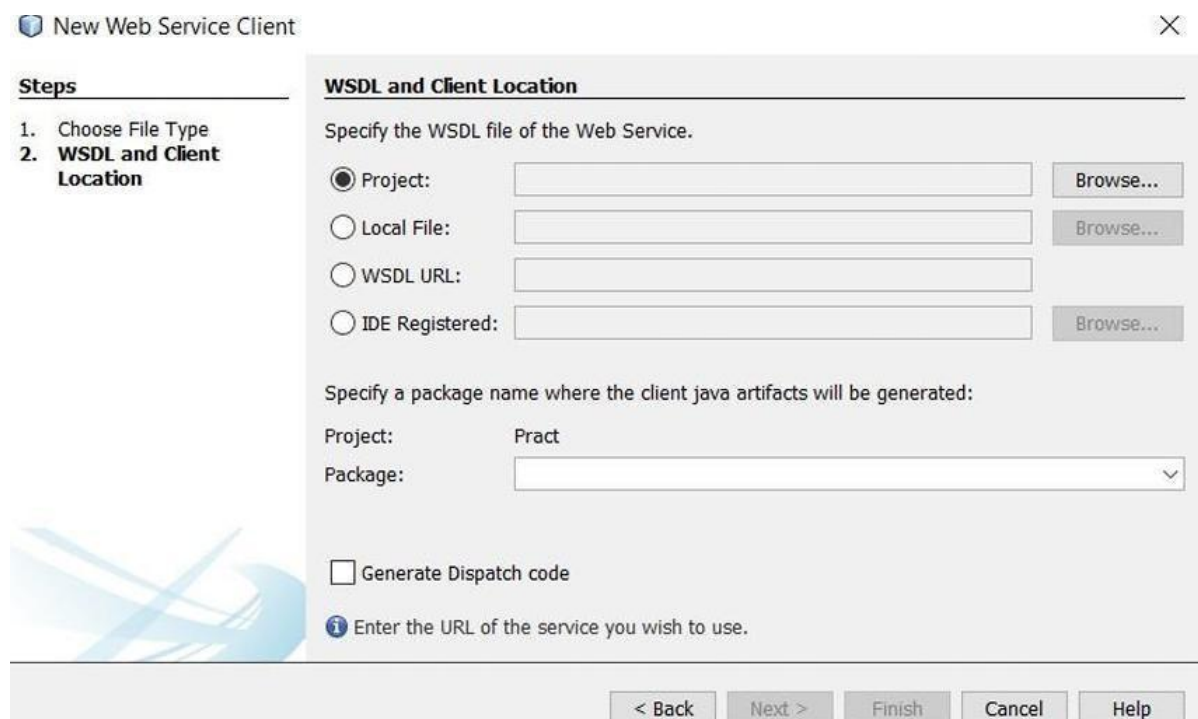
16] In input.jsp create a form for taking user input as shown below:


```
<!DOCTYPE html>
<html>
<head>
  <meta http-equiv="Content-Type" content="text/html; charset=UTF-8">
  <title>JSP Page</title>
</head>
<body>
  <form action="output.jsp">
    <pre>
      Enter the currency in rupees : <input type="text" name="t1">
      <input type="submit"> <input type="reset">
    </pre>
  </form>
</body>
</html>
```

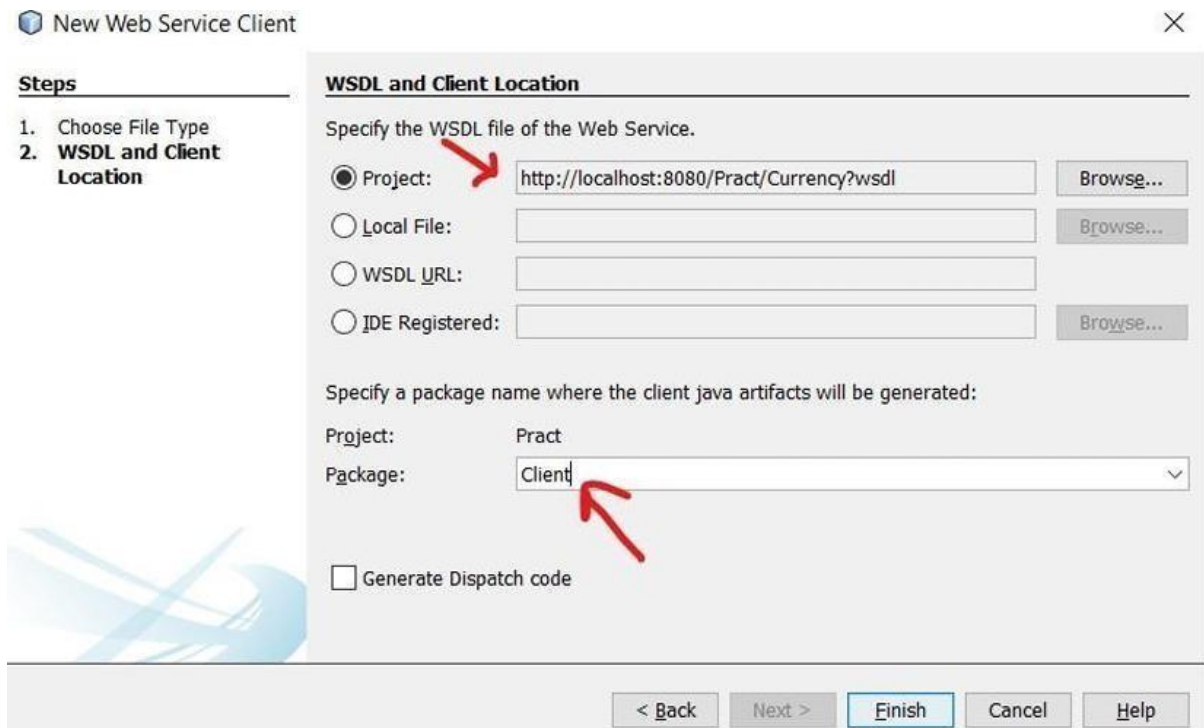


In this code set action = the jsp file where u want output and give name to textbox input.

17] now we have to create web service client, for same right click on project and select new then select web service client you will get the following screen:



At this step click on browse and select your project and click ok after clicking ok you will get wsdl url as shown below:

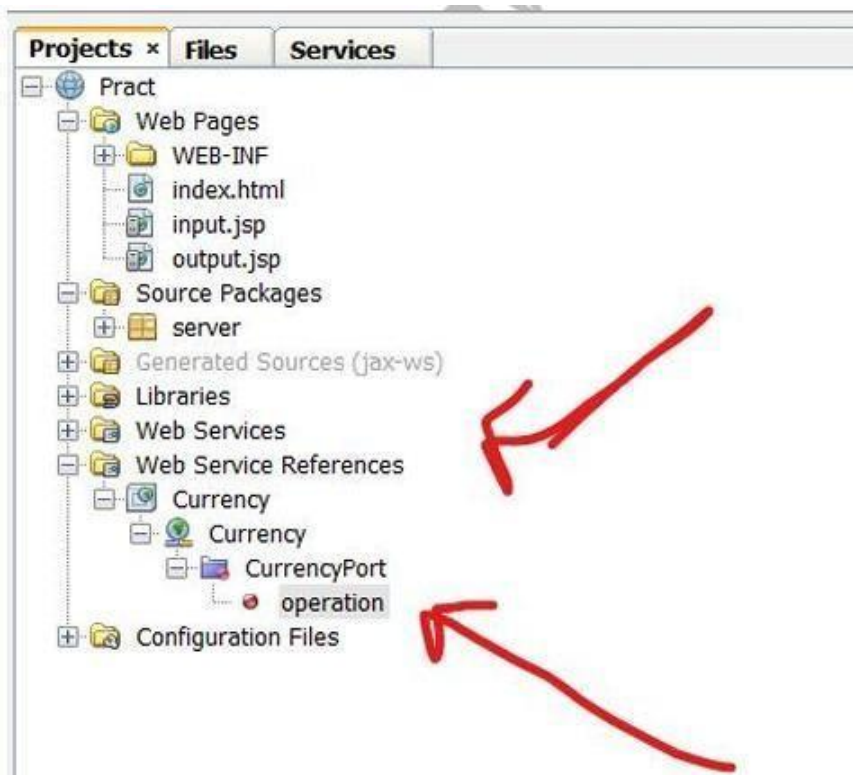


At this stage copy the url and paste it in notepad for future.

Give package name as "Client".

Now click on finish.

18] now in project section you can see a new folder is been created named "web servicereference". double click to expand it you should get the following



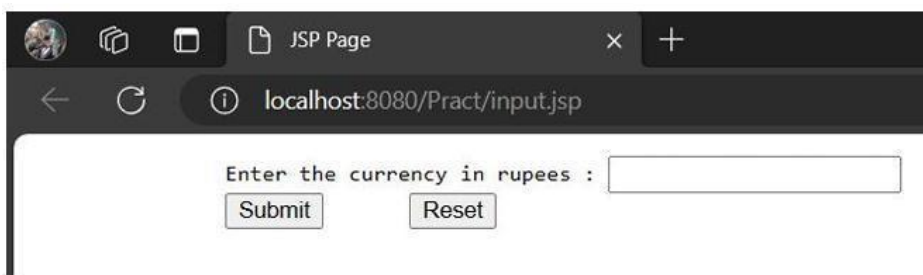
Hold the operation or your operation name and drag it into the output.jsp in body tag as shown :
You will get the following auto generated code:

```
<body>
    <!-- start web service invocation --><hr/>
    <%
    try {
        Client.Currency_Service service = new Client.Currency_Service();
        Client.Currency port = service.getCurrencyPort();
        // TODO initialize WS operation arguments here
        double a = 0.0d;
        // TODO process result here
        java.lang.String result = port.operation(a);
        out.println("Result = "+result);
    } catch (Exception ex) {
        // TODO handle custom exceptions here
    }
    %>
    <!-- end web service invocation --><hr/>
</body>
```

19] now make changes as shown below in the code:

```
<!-- start web service invocation --><hr/>
<%
try {
    Client.Currency_Service service = new Client.Currency_Service();
    Client.Currency port = service.getCurrencyPort();
    // TODO initialize WS operation arguments here
    double a = Double.parseDouble(request.getParameter("t1"));
    // TODO process result here
    java.lang.String result = port.operation(a);
    out.println(result);
} catch (Exception ex) {
    // TODO handle custom exceptions here
}
%>
<!-- end web service invocation --><hr/>
```

20] now Deploy our project again and right click on input.jsp and click run file you will redirect to a browser page as shown :



Enter any numerical value in text box and click on submit you will get the following output:



Python client:

1] for consuming java web services in python we should repeat the above steps till step np.19 . Note: for getting output the project or web service should be deployed . After doing all of the steps write the following code in python idle:

```
from zeep import Client wsdl_url='http://localhost:8080/pract/Currency?wsdl'
client=Client(wsdl_url) #call the method on the webservice d=float(input("Enter Currency in Indian Rupees:")) result = client.service.InrtoDollar(d) print(result)
```

2] replace wsdl_url with the url which was copied at the time of web service client creation .

```
*2.py - C:\Users\LENOVO\AppData\Local\Programs\Python\Python311\2.py (3.11.4)*
File Edit Format Run Options Window Help
from zeep import Client
wsdl_url='http://localhost:8080/pract/Currency?wsdl'
client=Client(wsdl_url)
#call the method on the webservice
d=float(input("Enter Currency in Indian Rupees:"))
result = client.service.InrtoDollar(d)
print(result)
```

4] after running the above code you should get the following output:

```
= RESTART: C:\Users\LENOVO\AppData\Local\Programs\Python\Python311\2.py
Enter Currency in Indian Rupees:12000
The Indian rupees 12000.0 in Dollars is 144.28279427678248
```

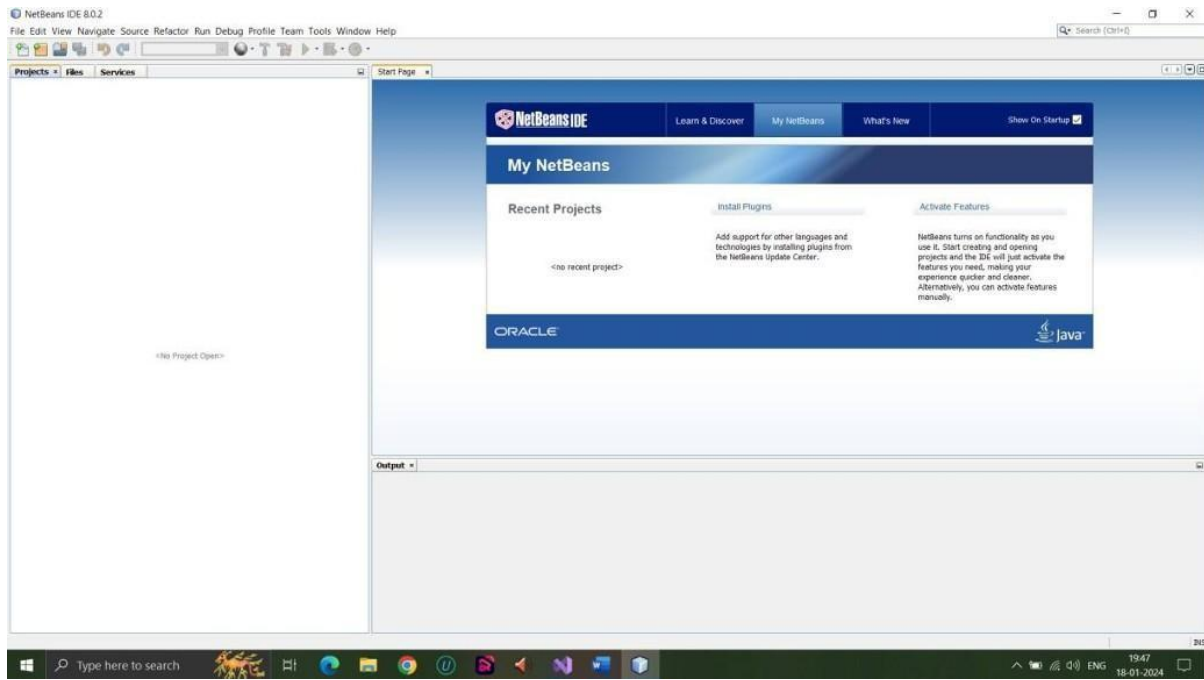
Conclusion: Hence, we have successfully created and executed the conversion of INR to Dollar using simple web services and called from different platforms like JAVA and PYTHON.

Practical 2

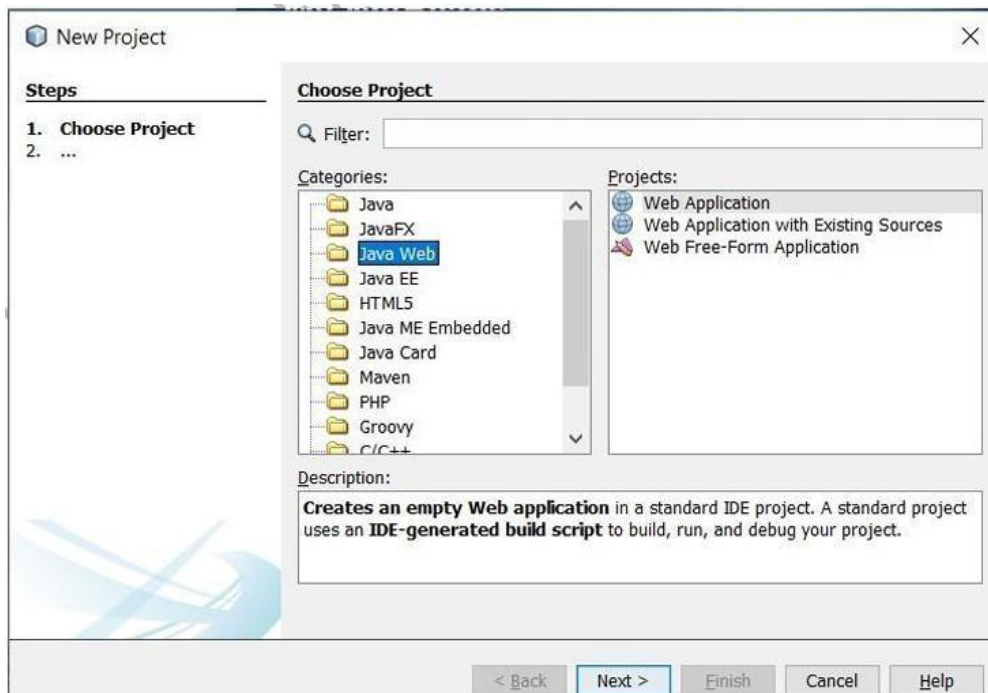
Aim: Create a Simple SOAP service.

Steps:

1] Open the NetBeans , and you will get the following screen. Close the startpage.



2] Now click on the file tab and click on new project you will get the following screen:



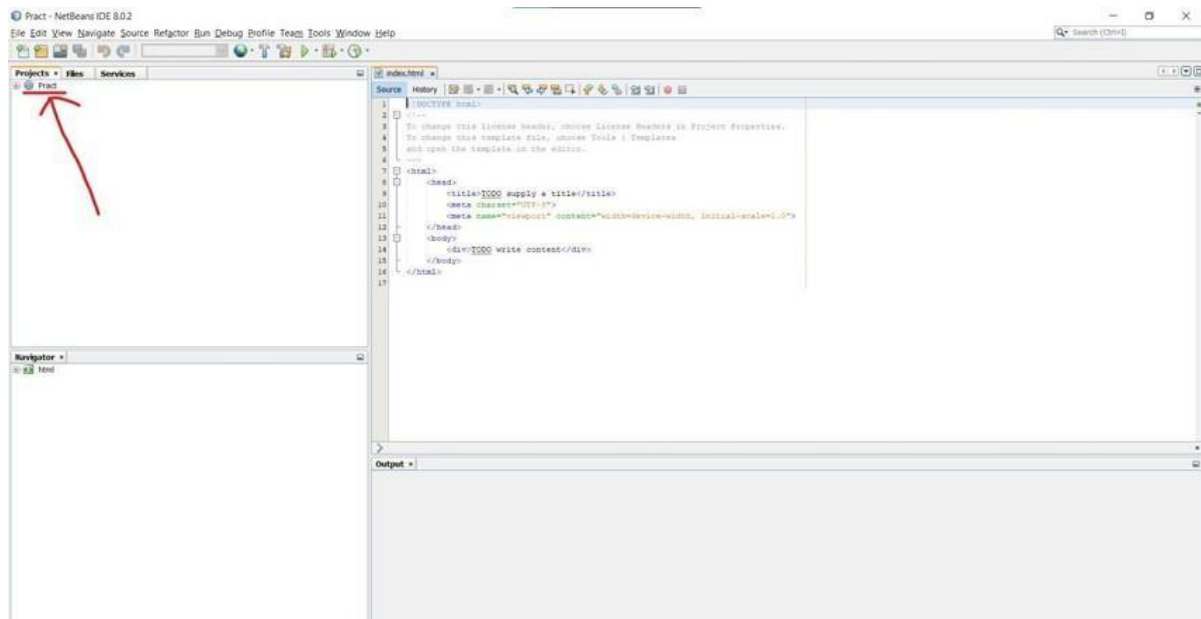
3] In Categories select Java Web and in Projects , Select Web Application .After selecting click on next . You will get the following window:

The screenshot shows the 'New Web Application' dialog box with the 'Name and Location' step selected. The 'Steps' list on the left includes: 1. Choose Project, 2. **Name and Location**, 3. Server and Settings, and 4. Frameworks. The 'Name and Location' section contains the following fields: 'Project Name' with the value 'Pract1', 'Project Location' with the value 'C:\Users\LENOVO\Documents\NetBeansProjects' and a 'Browse...' button, and 'Project Folder' with the value 'C:\Users\LENOVO\Documents\NetBeansProjects\Pract1'. Below these fields is a checkbox labeled 'Use Dedicated Folder for Storing Libraries' which is unchecked, followed by a 'Libraries Folder' field and a 'Browse...' button. A note states: 'Different users and projects can share the same compilation libraries (see Help for details)'. At the bottom, a red error message reads: 'Project Folder already exists and is not empty.' The bottom of the dialog has buttons for '< Back', 'Next >', 'Finish', 'Cancel', and 'Help'.

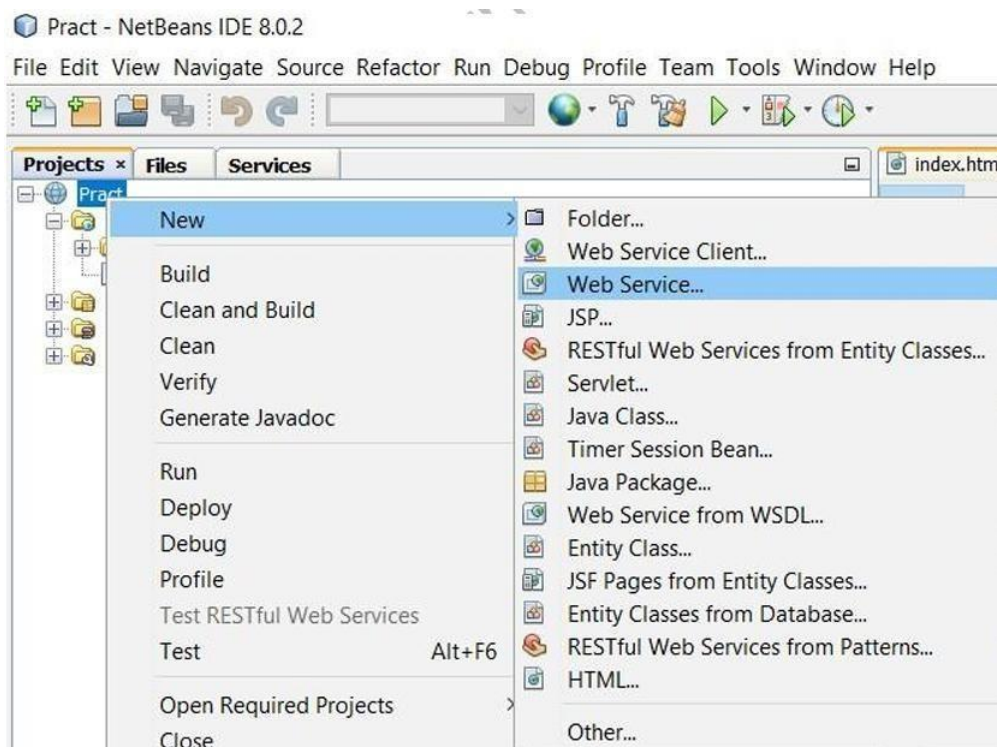
4] Now Give name to the Project Name: , and click on next . you will get the following window then again click on finish:

The screenshot shows the 'New Web Application' dialog box with the 'Server and Settings' step selected. The 'Steps' list on the left includes: 1. Choose Project, 2. Name and Location, 3. **Server and Settings**, and 4. Frameworks. The 'Server and Settings' section contains the following fields: 'Add to Enterprise Application' with a dropdown menu showing '<None>', 'Server' with a dropdown menu showing 'GlassFish Server 4.1' and an 'Add...' button, 'Java EE Version' with a dropdown menu showing 'Java EE 7 Web', and 'Context Path' with the value '/Pract'. The bottom of the dialog has buttons for '< Back', 'Next >', 'Finish', 'Cancel', and 'Help'.

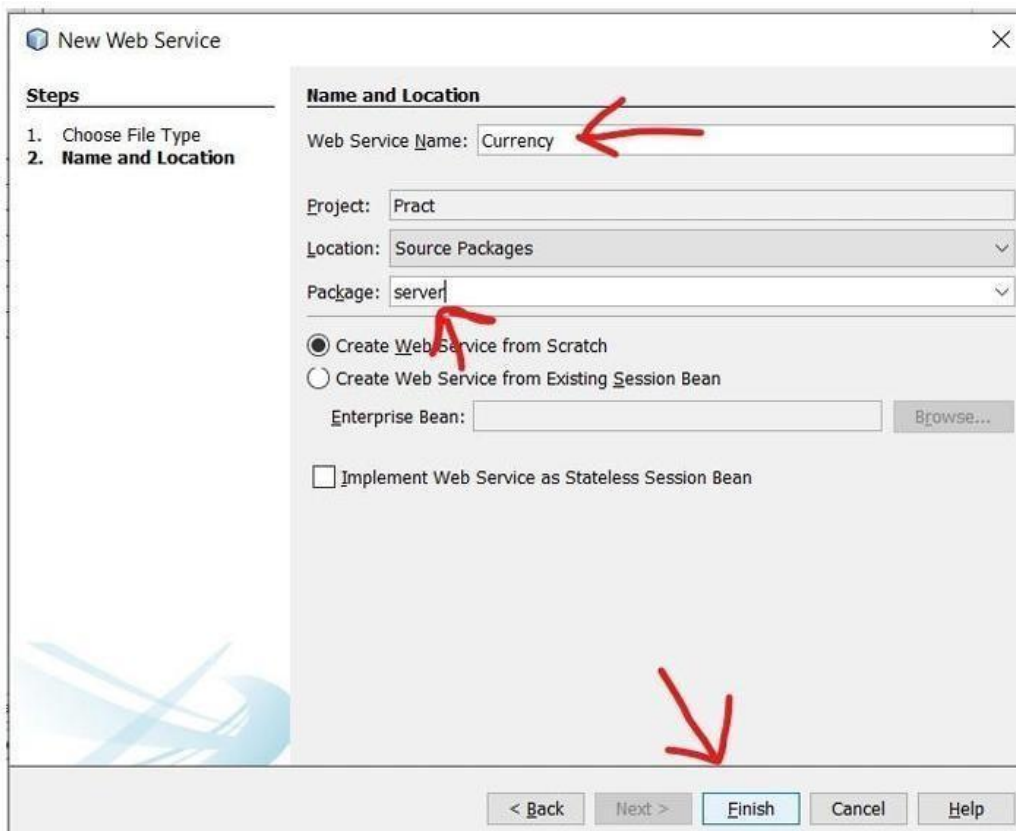
5] You will get the following screen now carefully see in projects section your recently created project appears double click to expand it:



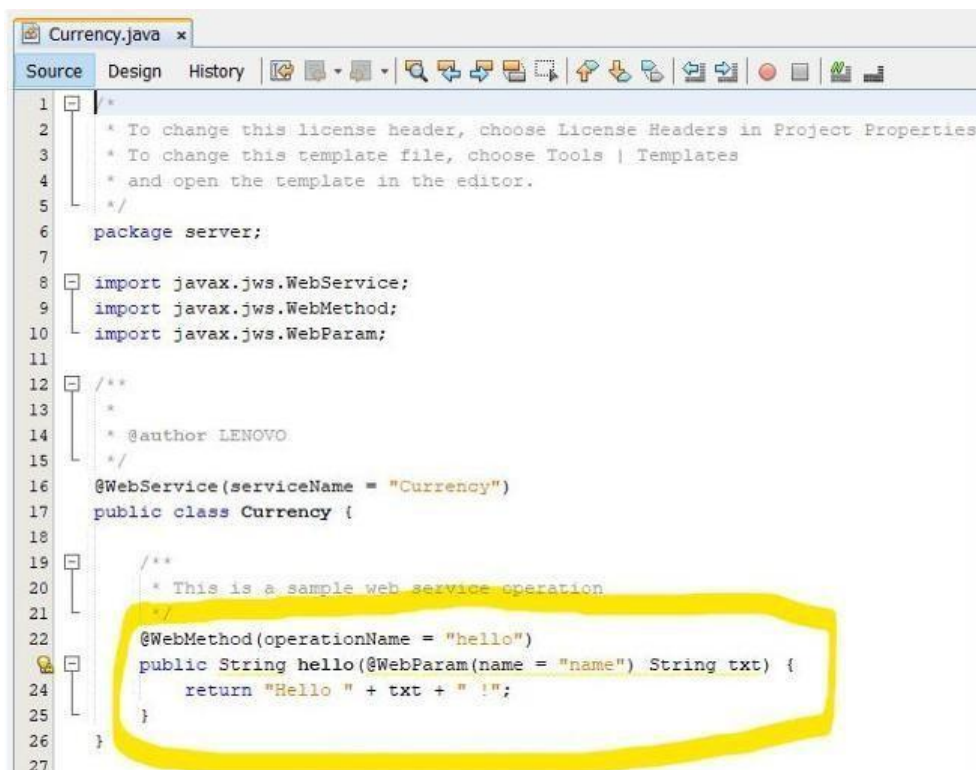
6] Now Right click on the project and select new and then select Web Service ,As shownBelow:



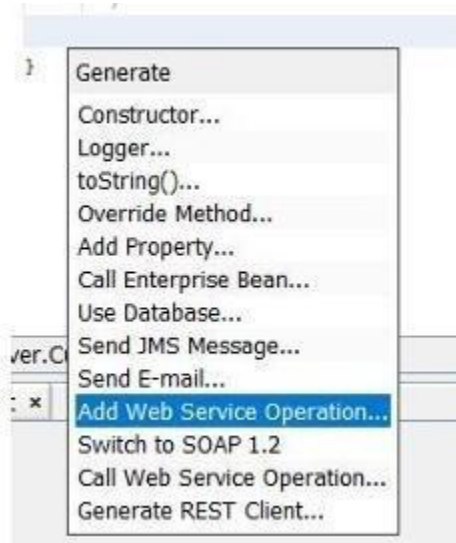
7] after clicking Web Service following window should appear , now give name to web service and give package as "server". As shown in the image:



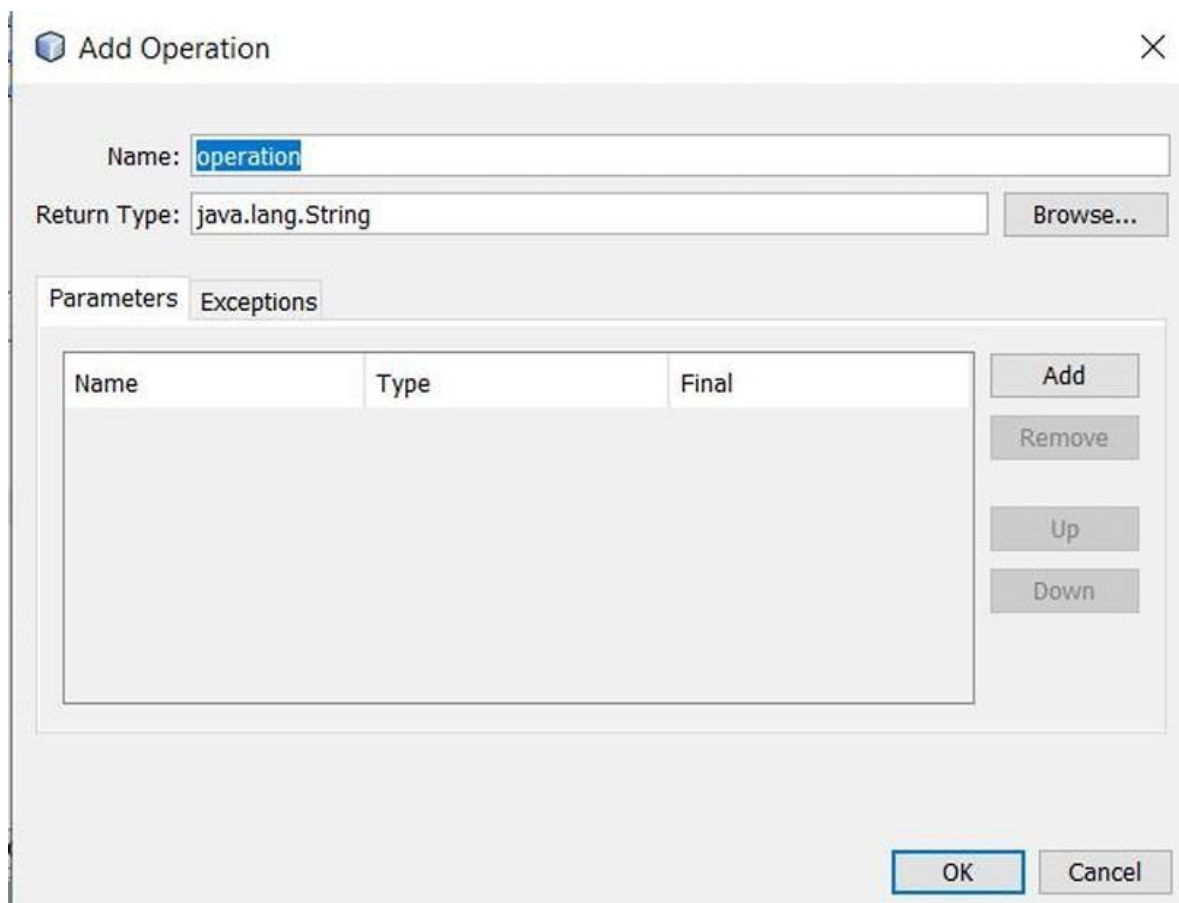
After clicking finish you should get the following window erase the mentioned code :



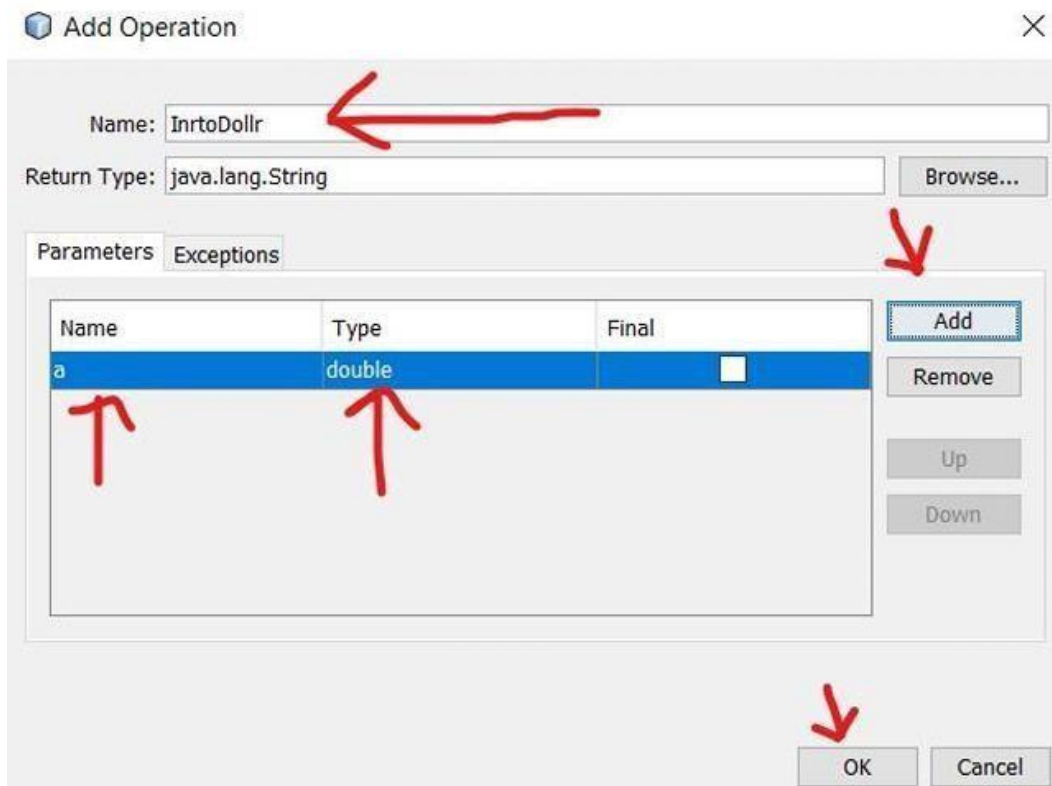
8] Nowright click any where and click on insert code and select Add Web Service Operation:



9] the following window would appear: Just give name to the method or operation and click on add button to add parameters to the method as here we are converting dollar to rupees we should need only one parameter .



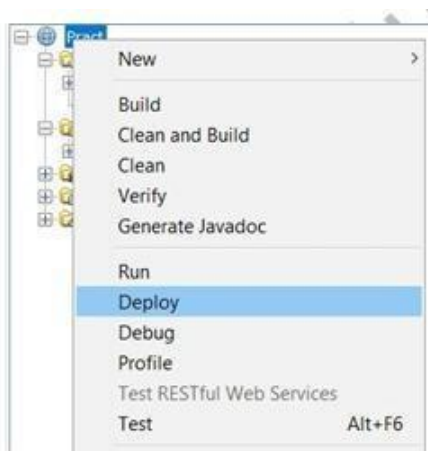
10] give the name to the variable select data type as double and click on ok as shown below:



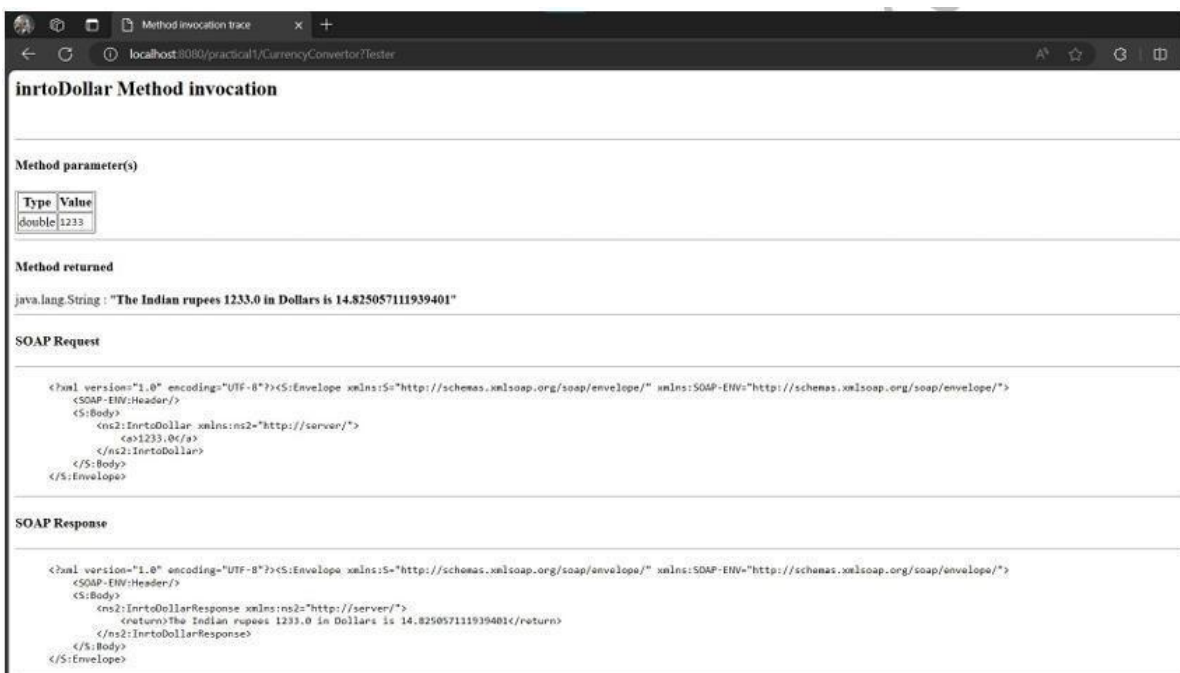
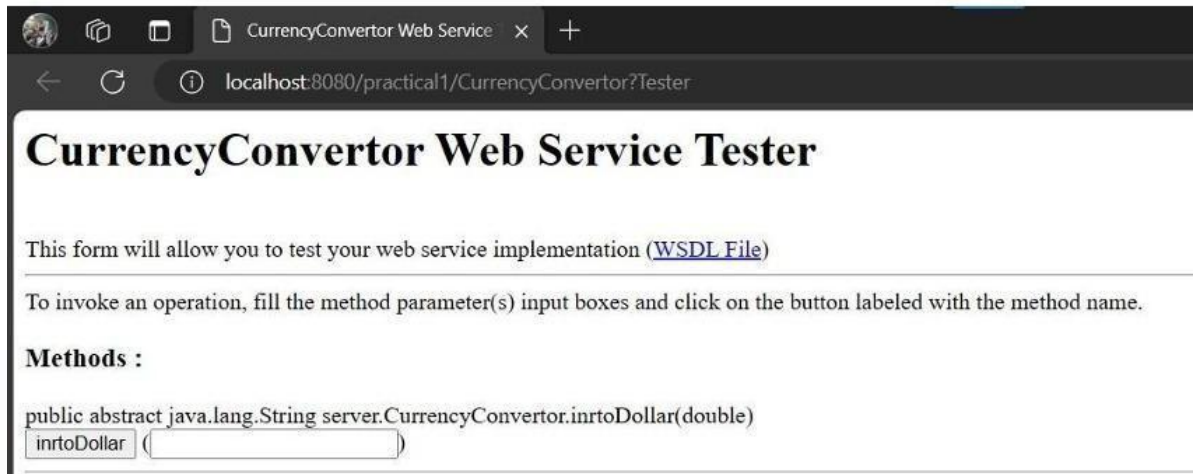
11] After clicking ok code will autogenerate , make changes In that code as mentioned below:

```
@WebMethod(operationName = "InrtoDollar")
public String InrtoDollar(@WebParam(name = "a") double a) {
    //TODO write your implementation code here:
    return "The Indian rupees "+a+" in Dollars is "+(a/83.17);
}
```

12] now our webservice is ready now right click on project and click on deploy:



13] now right click on webservice and click on test web service you will get the following output:



So this is how we created our web service and deployed it.

14] Now our web service is successfully deployed. Right click on web pages and select new and select jsp as shown below:

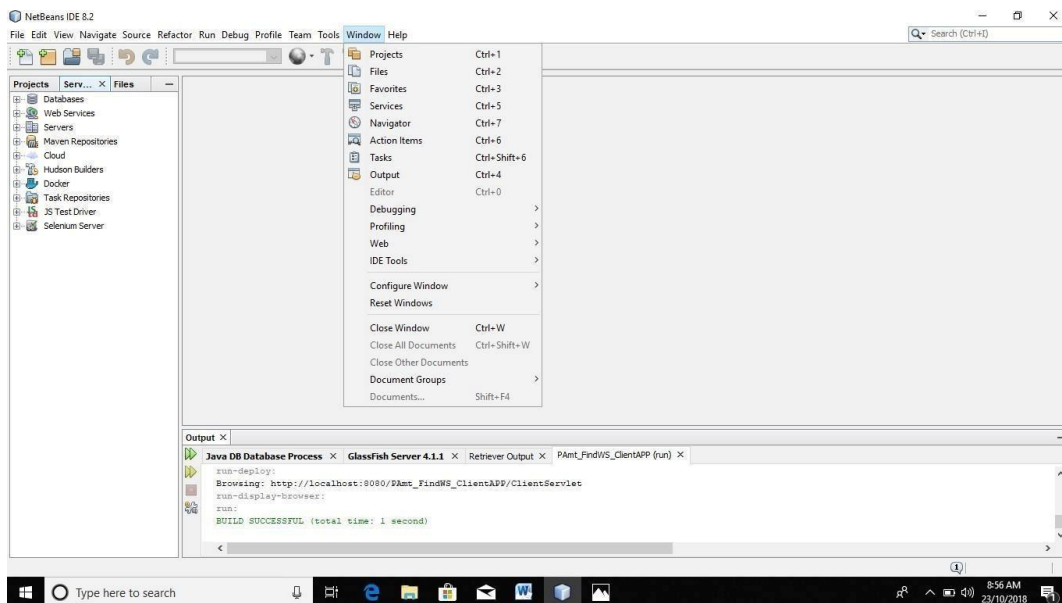
Conclusion: Hence, we have successfully created SOAP web services.

Practical 3

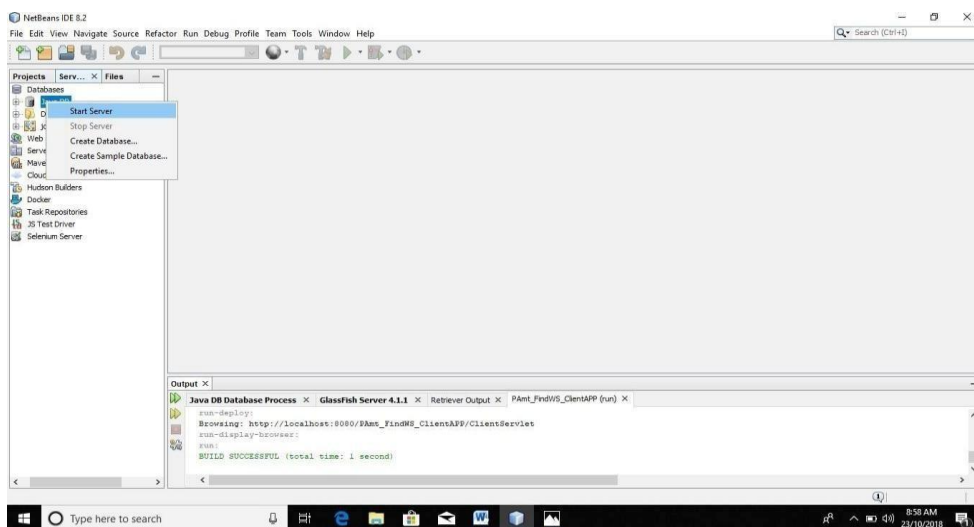
Aim: Create a Simple REST Service to demonstrate CRUD Operations with “Student” database.

Steps:

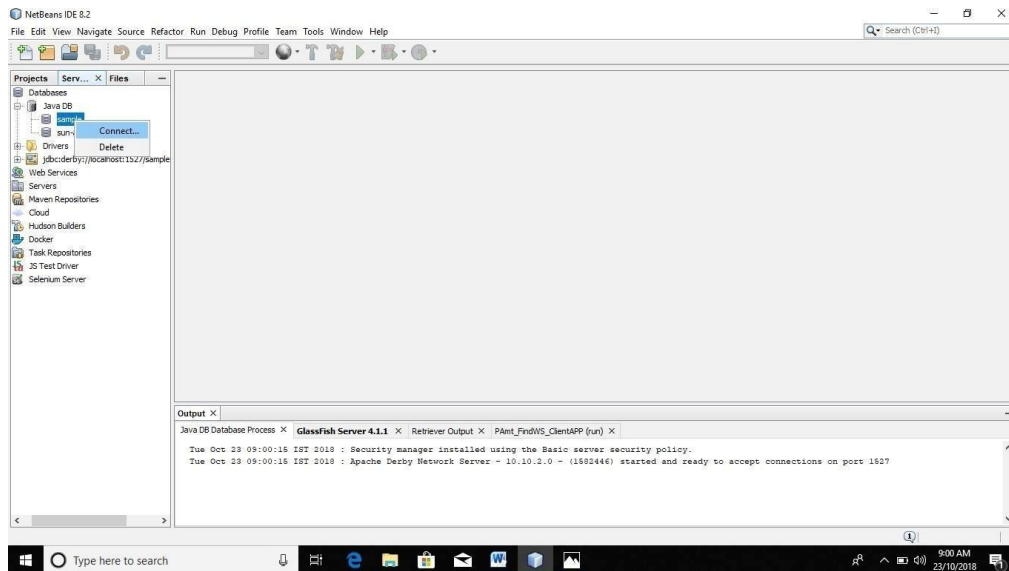
1. Click on Window menu and click on Projects, Files & Services to open it.



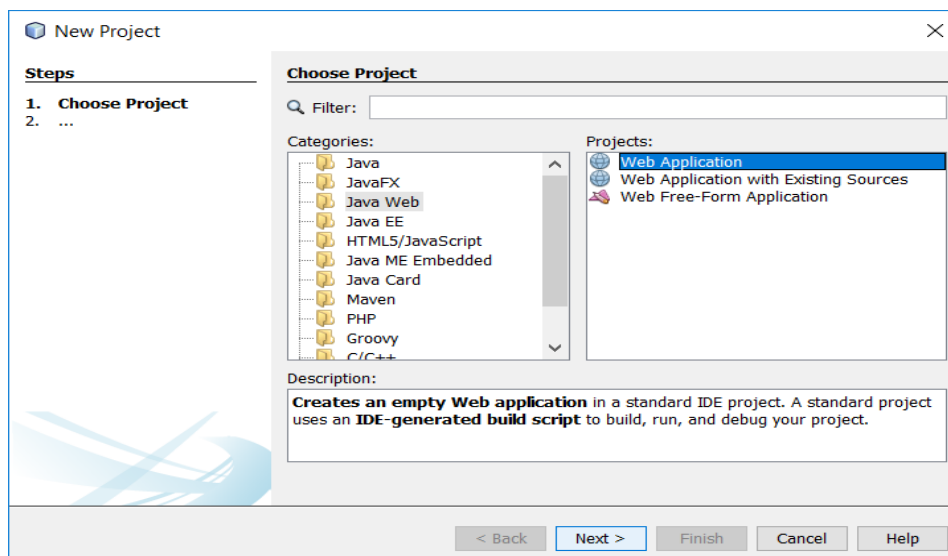
2. Right click on Java DB and then click on Start Server to start the server .

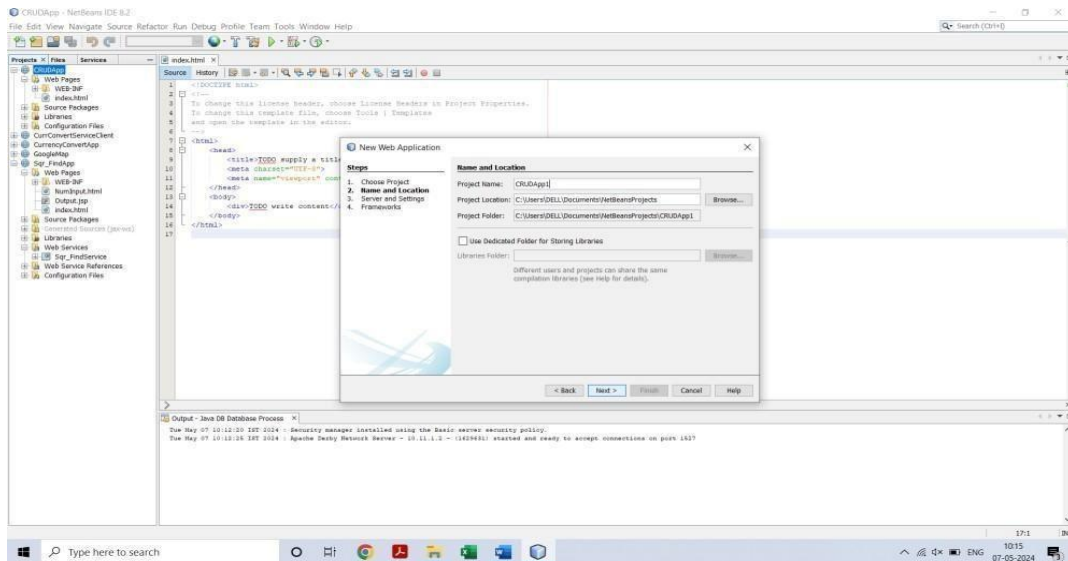


3. Now expand Java DB and right click on sample and then click on connect to connect the sample database with server.

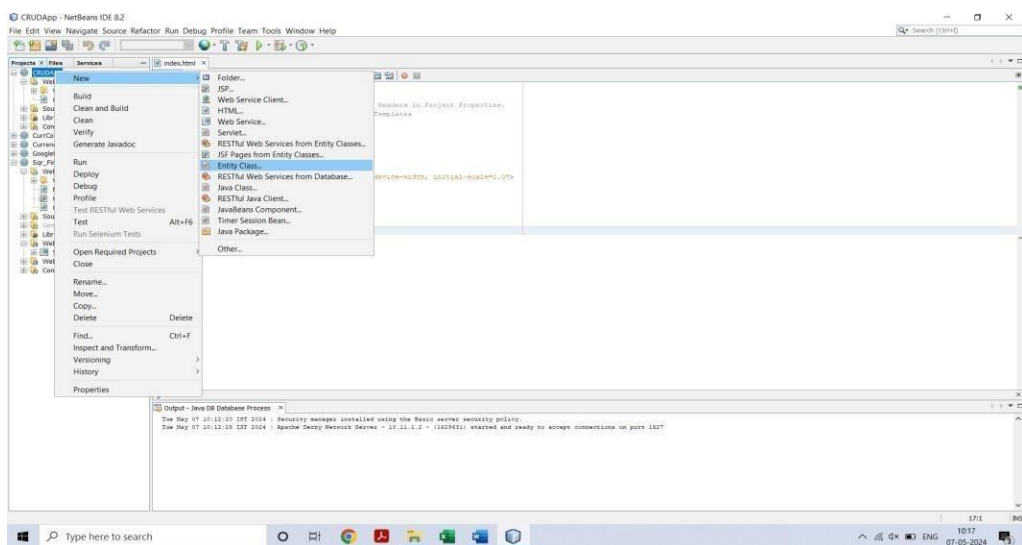


4. Now create a web application with the name CRUD_Operation. A window will open like following pic.



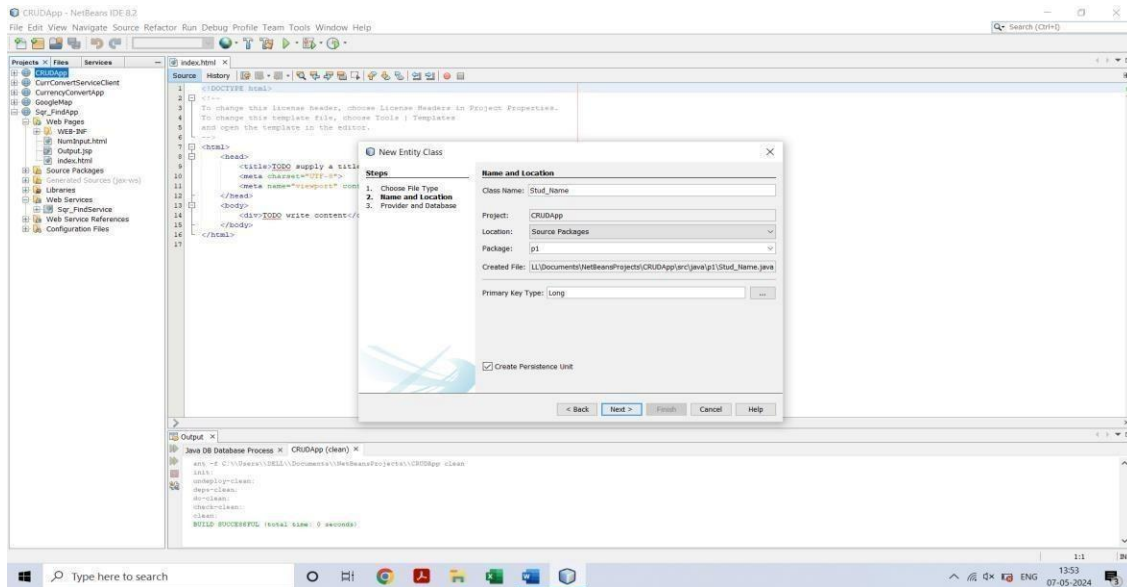


5. Create an entity class. Right click on project name -> New -> Entity Class.



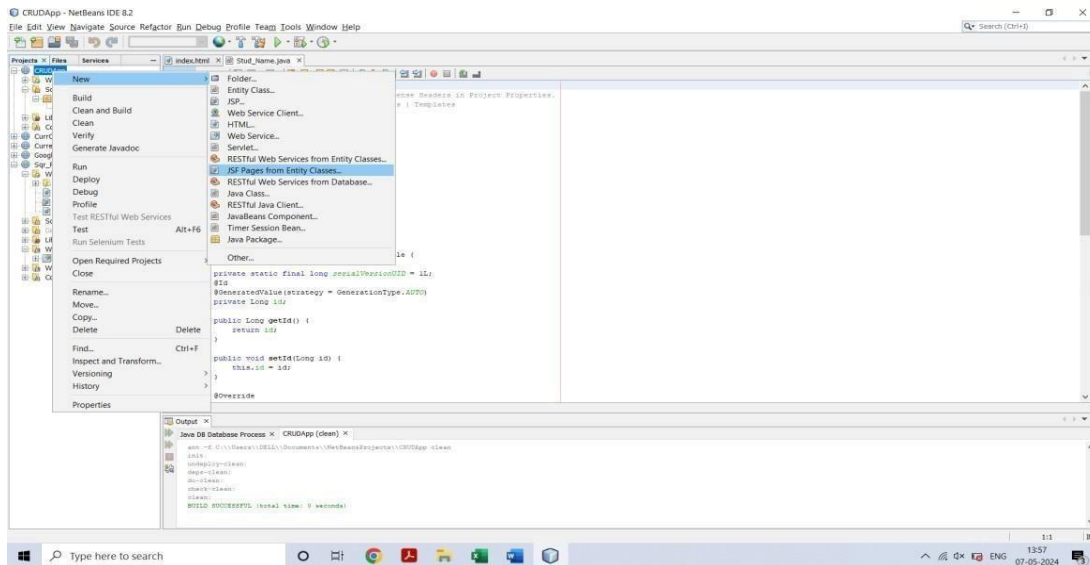
6. A window will appear like bellow pic. Enter following data and click on Next Class Name -> Stud_Name Package Name -> p1

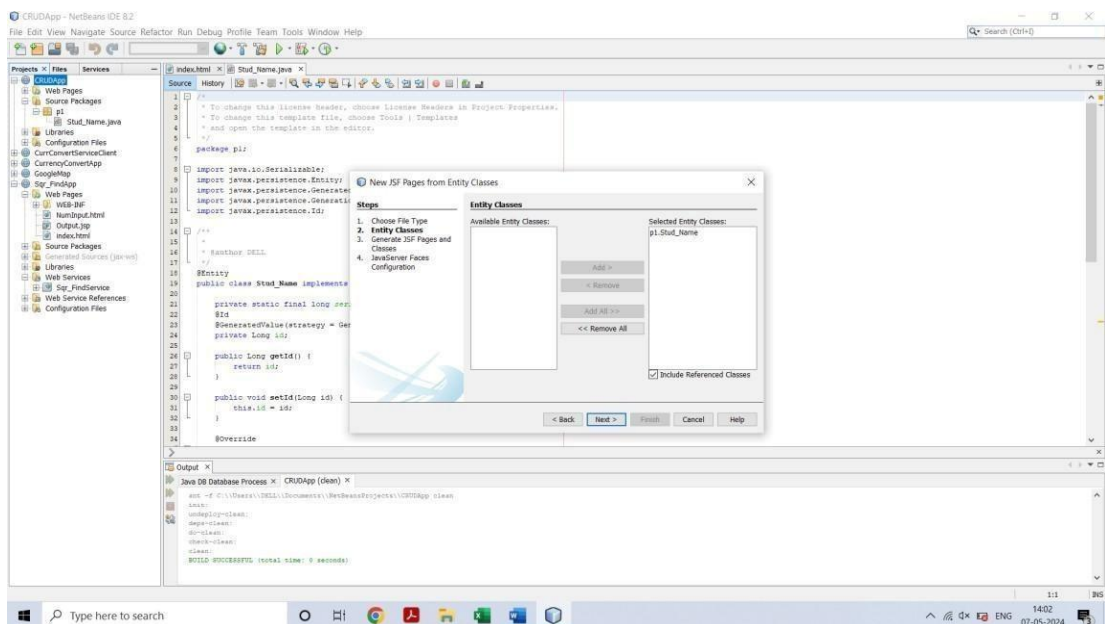
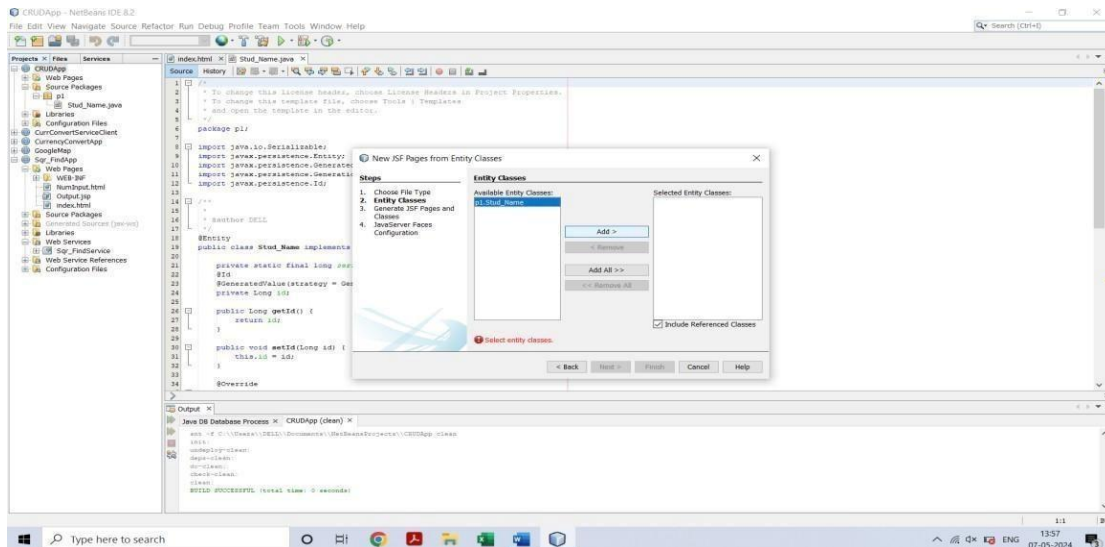
7. Click on finish.



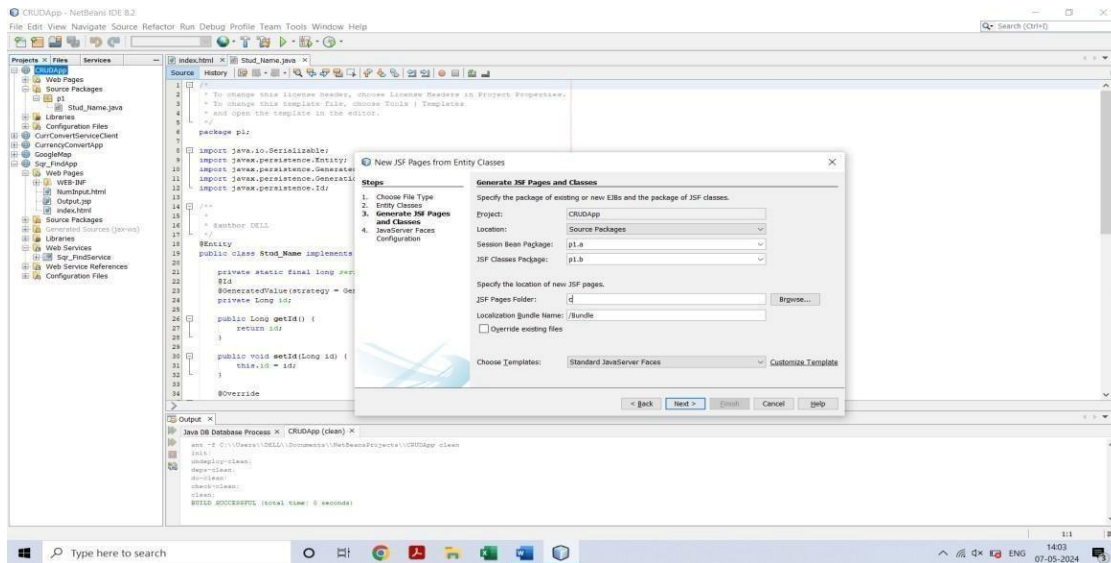
8. Right click on project name and create JSF Pages from Entity Classes.

9. Select and click on Add button and then Next button on below

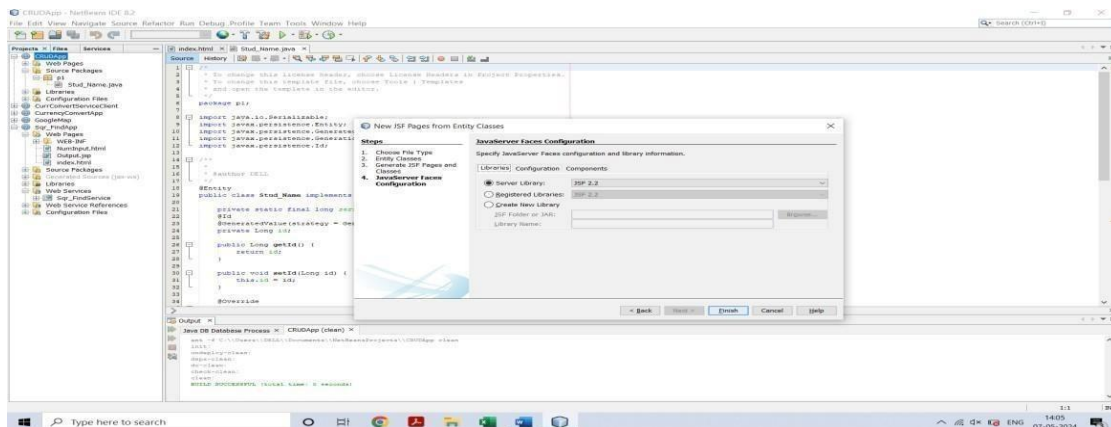




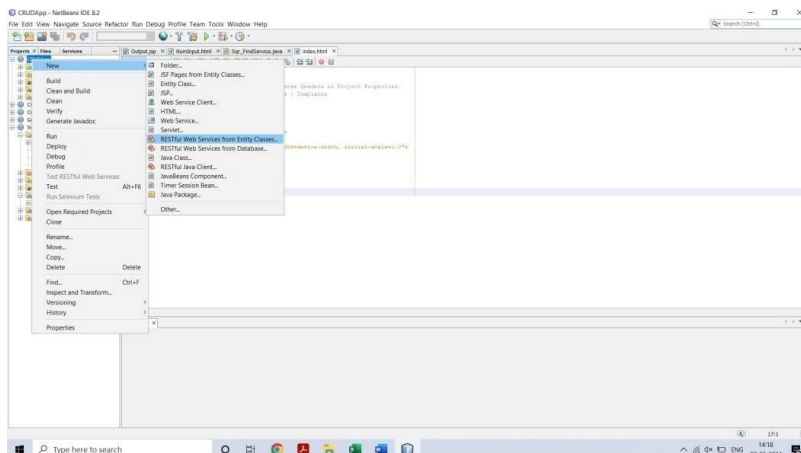
10. A window like below will appear on the screen. Enter the data into that window as entered in below pic and click on Next button.

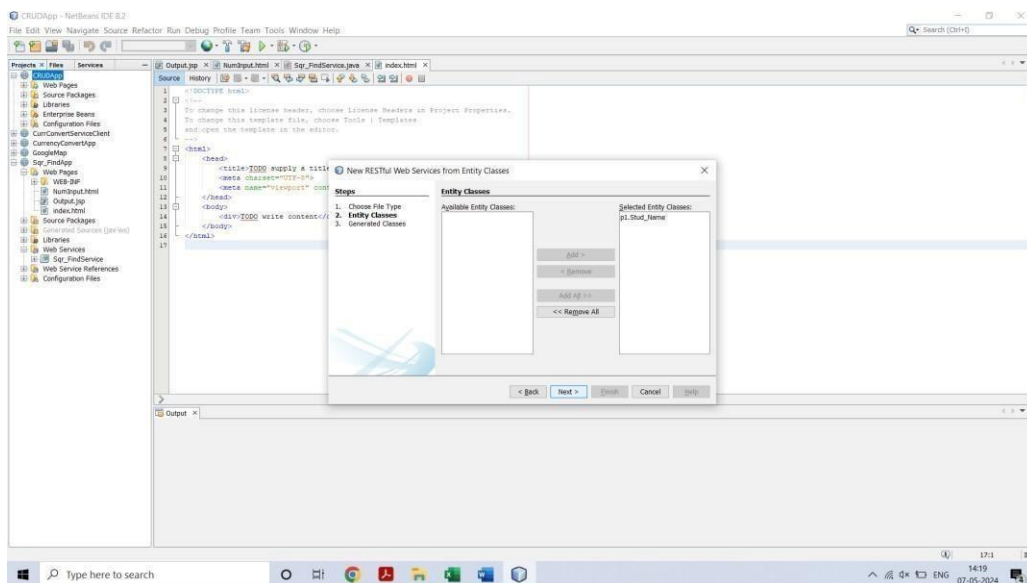
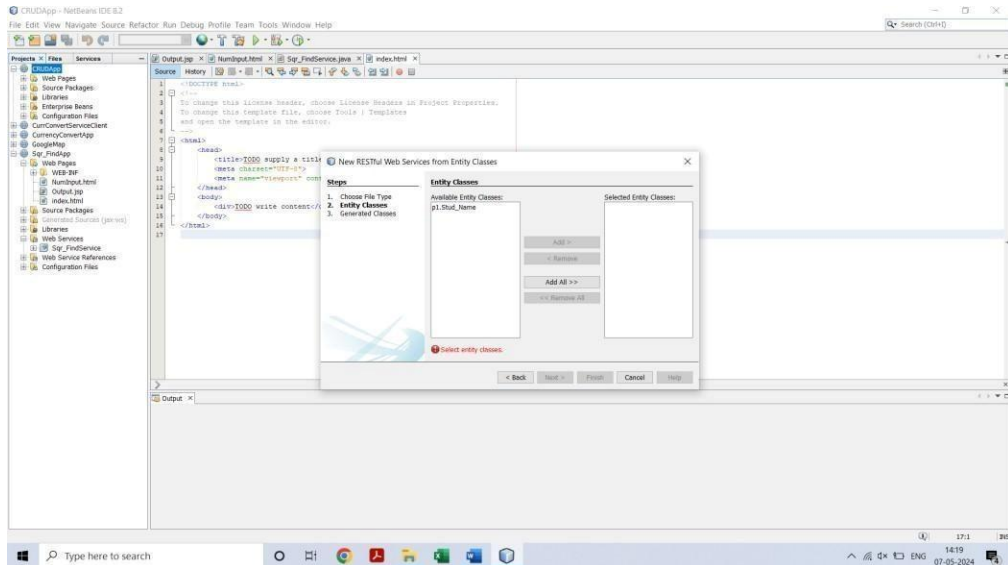


11. Now click on Finish.

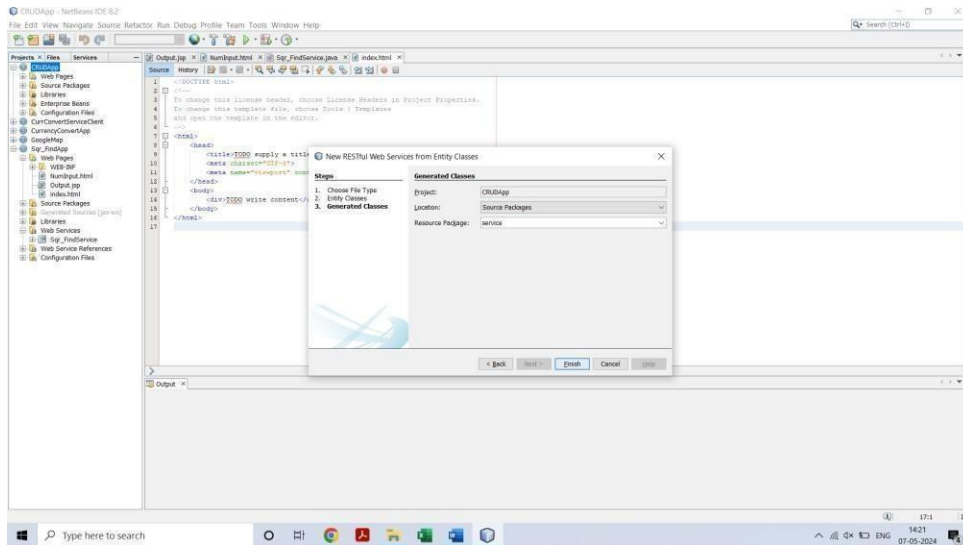


12. Right click on project name and create RESTful Web Services from Entity Classes.

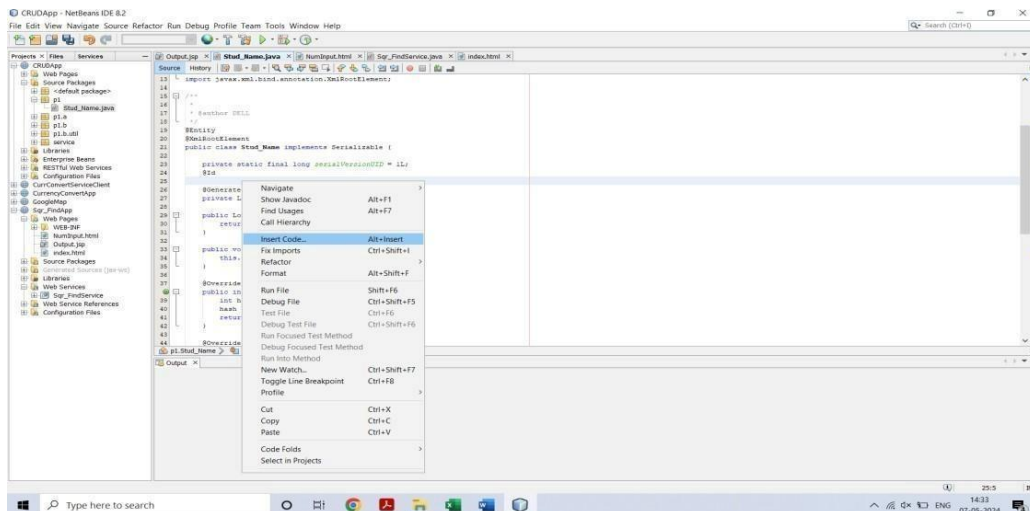


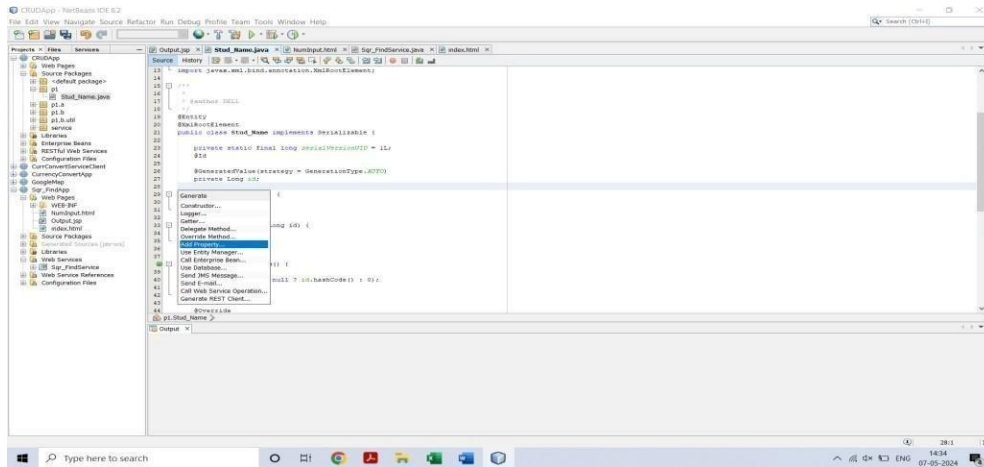


13. Repeat step 9 and then it will go on next page. Then enter the p1.service in Resource Package and then click on Finish button.

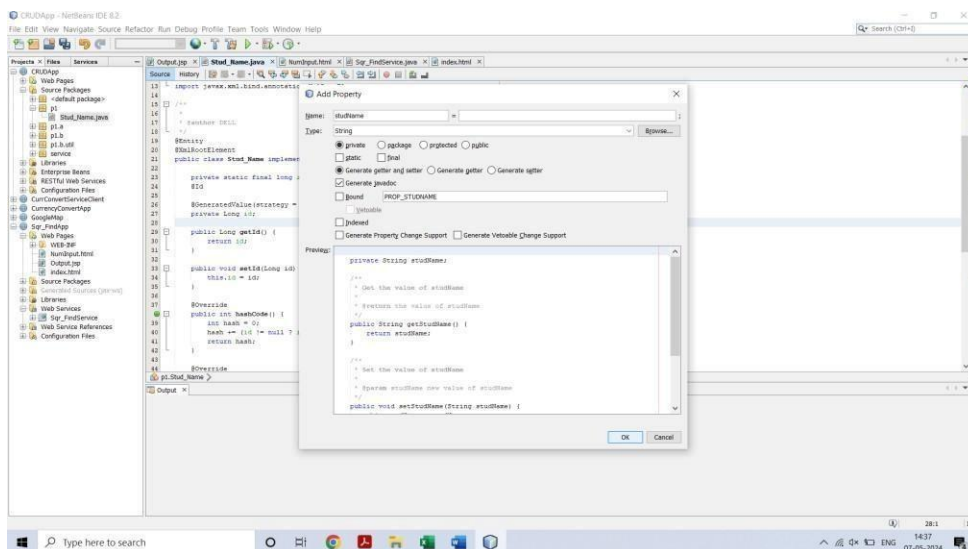


14. Now open Stud_Name.java file under p1 package.
15. In this file at line number 24, do the right click and select Insert Code.
16. A new list will appear. Click on Add Property

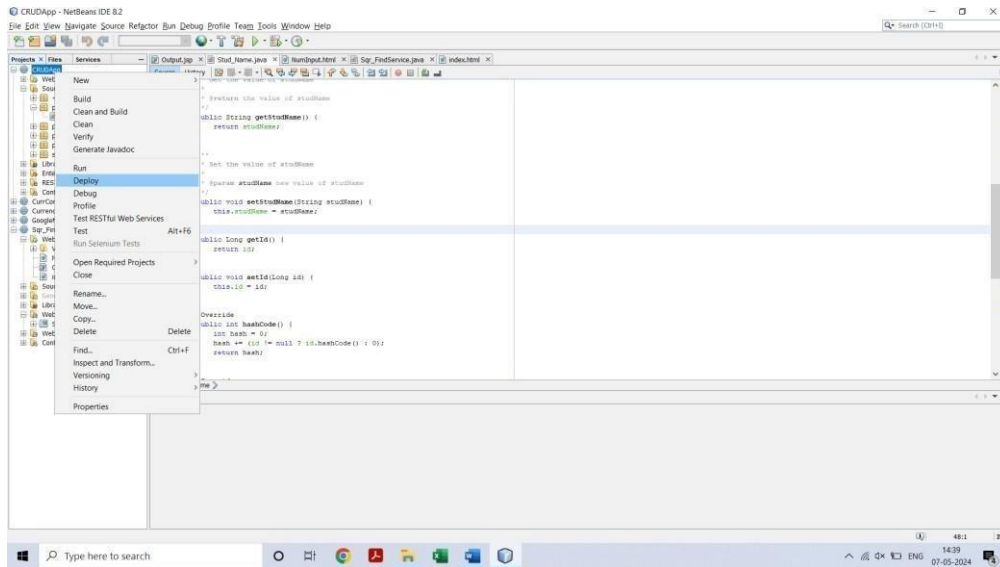




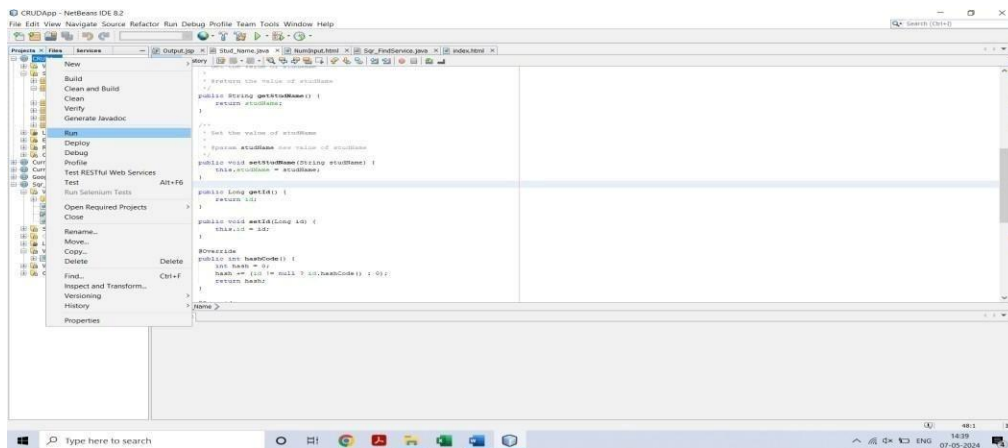
17. A new window will open. Enter name as studName. Make sure name should be exact same as of mine and then click on OK button. Actually we are setting getter and setter method for studName.



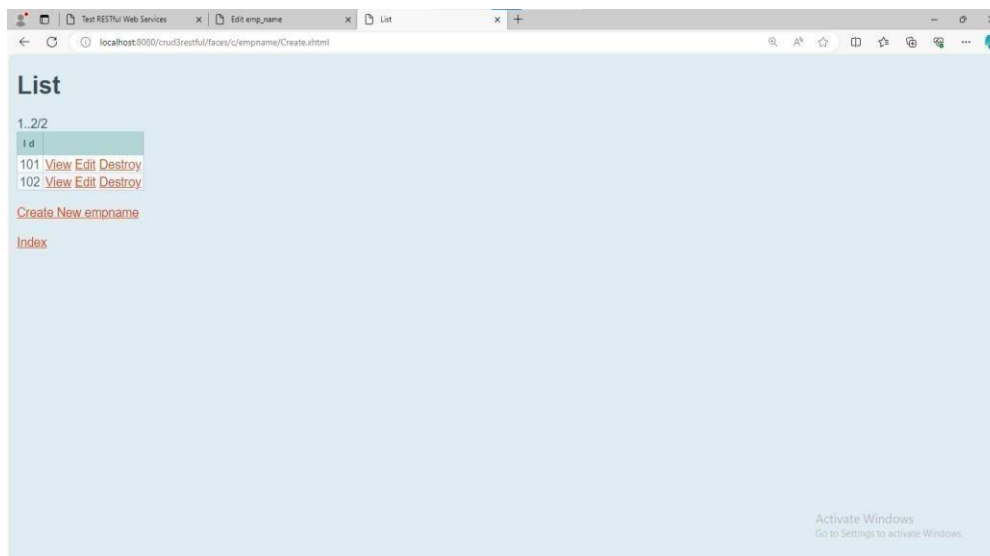
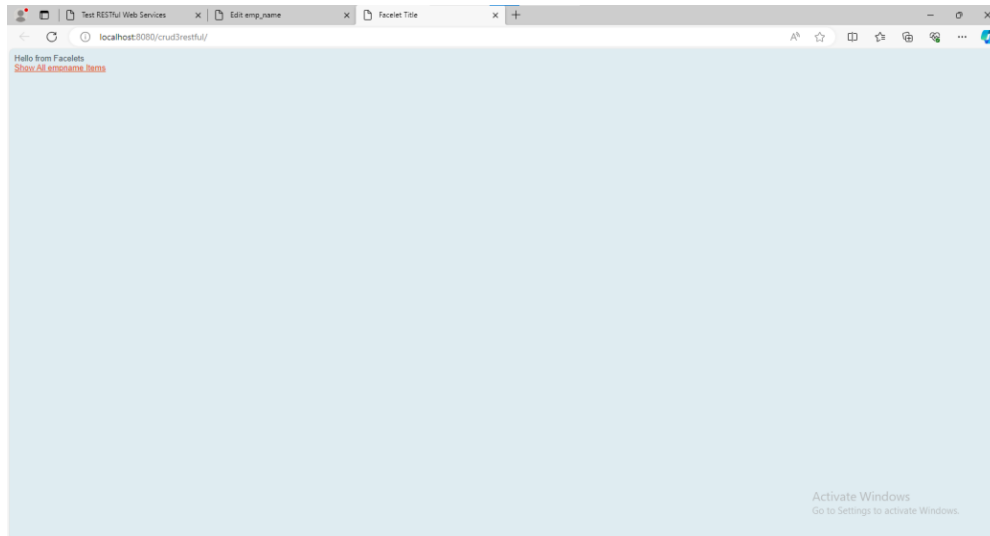
18. Now right click on web application name and deploy it.



19. Now right click on project name and run it



20. A window will open in browser like below....



Conclusion: Hence, we have successfully created restful web services.

Practical 4

AIM: Develop application to consume Google's search / Google's Map RESTful Web service.

Steps:

- Develop application to consume Google's search / Google's Map RESTful Web service.
- 1. First, we need to create a Java Web Application with any name, let it be GoogleMap here using Netbeans IDE.
- 2. The code inside the input.jsp will be similar to this input.jsp Input.jsp

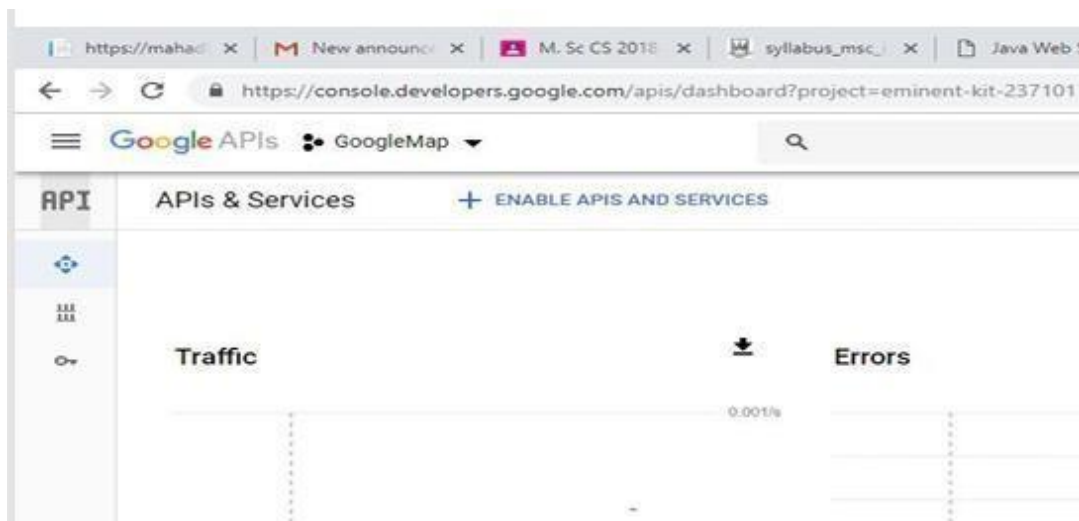
```
<%@pagecontentType="text/html" pageEncoding="UTF-8"%>
<!DOCTYPE html>
<html>
<head>
<meta http-equiv="Content-Type" content="text/html; charset=UTF-8">
<title>JSP Page</title>
</head>
<body>
<form action="index.jsp">
<pre>
Enter latitude:<input type="text" name="t1" />
Enter longitude:<input type="text" name="t2" />
<input type="submit" value="Show" />
</pre>
</form>
</body>
</html>
```

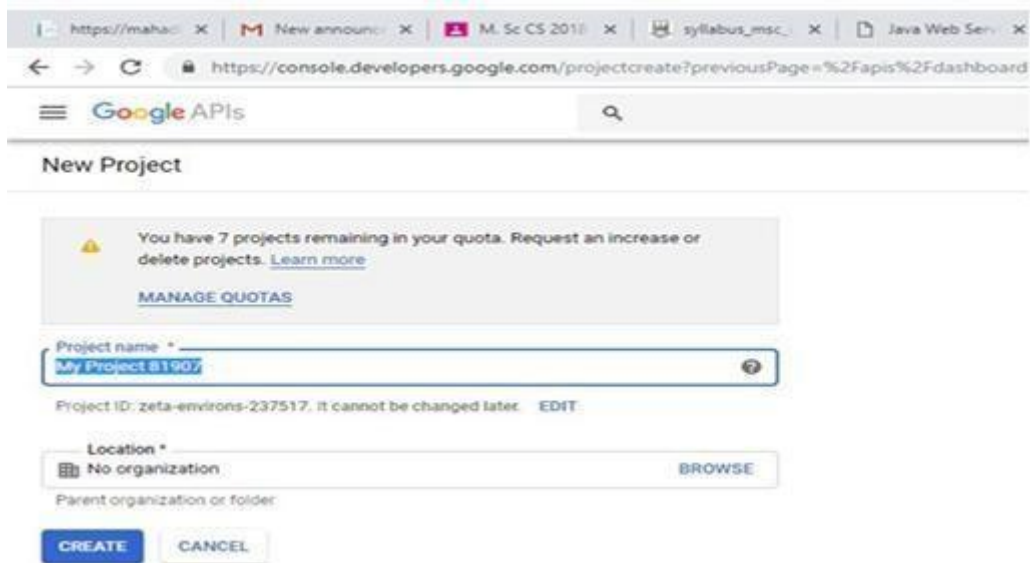
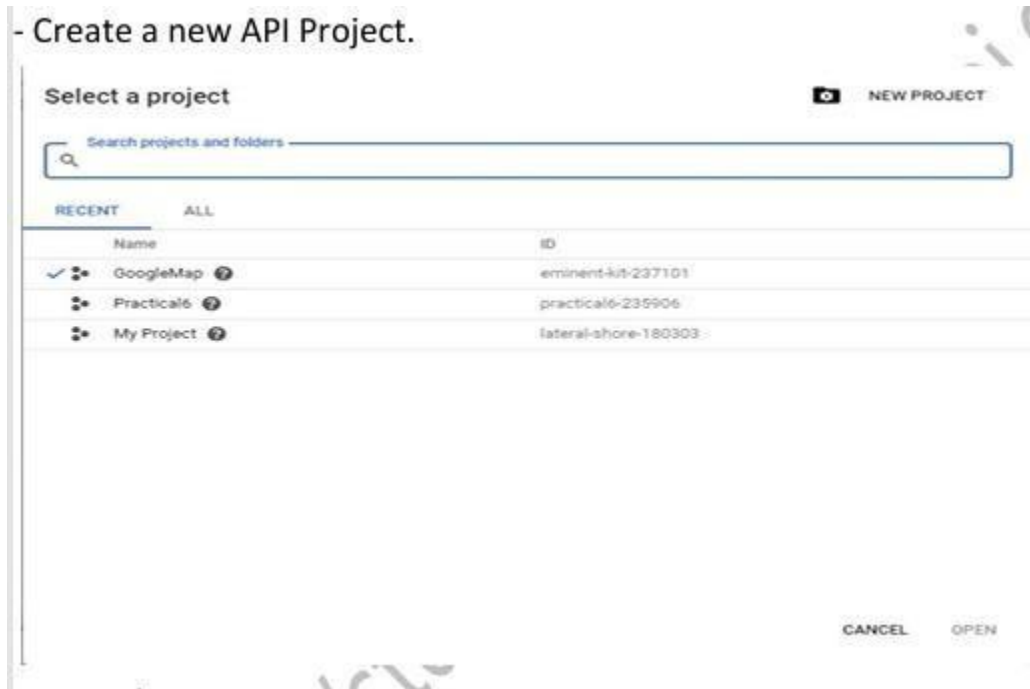
3. Before running the application, we need the Google API key. The steps are shown here:- Visit Google APIs Console (<https://console.developers.google.com>, you have to login with your Google account).
4. Before running the application, we need the Google API key. The steps are shown here:- Visit Google APIs Console (<https://console.developers.google.com>, you have to login with your Google account).

Follow these steps

1. Firstly Go to Google Maps - <https://console.developers.google.com>
2. Sign In with your Google account and agree to the terms and conditions
3. Create a project and name it as you want
4. Select the project and click on it
5. Navigate to APIs and Services and click on Libraries
6. Enable the following Libraries one by one
 1. Google Javascript API
 2. Google Place API
 3. Google Direction API
 4. Google Geolocation API
 5. Google Geocoding API
7. Navigate to APIs, Services, and click on Credentials
8. Click on Create Credentials and then the API key and it will show up a screen with API Key
9. Now you can optionally restrict the API key.

Almost done!





5. 4. Create another file index.jsp

Index.jsp

```
<%@pagecontentType="text/html" pageEncoding="UTF-8"%>
```

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

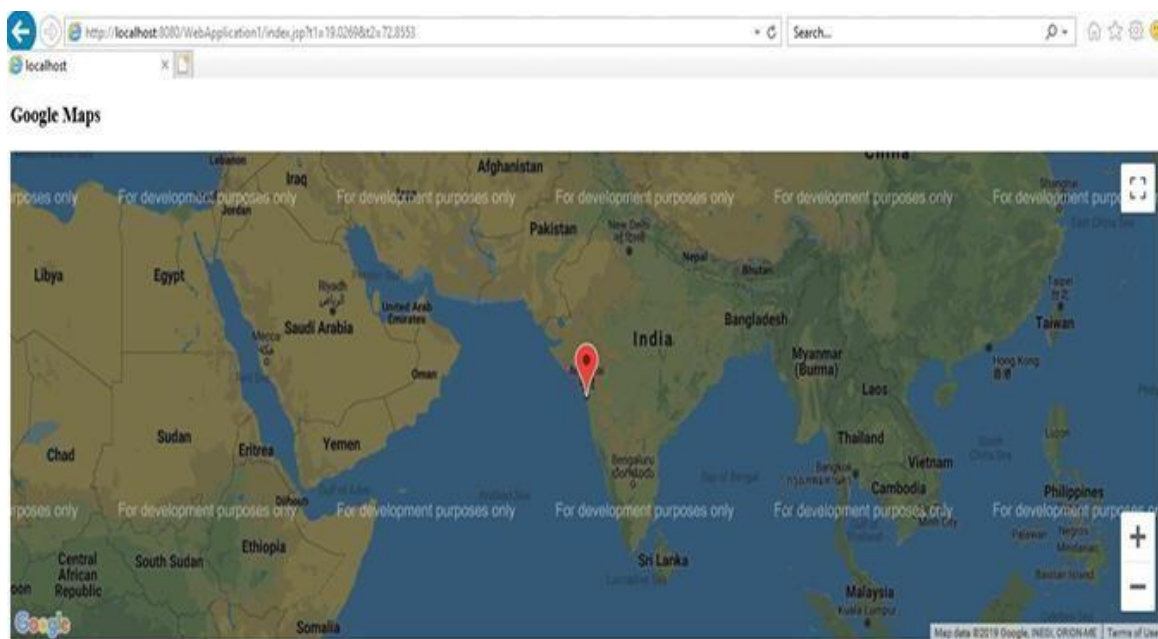
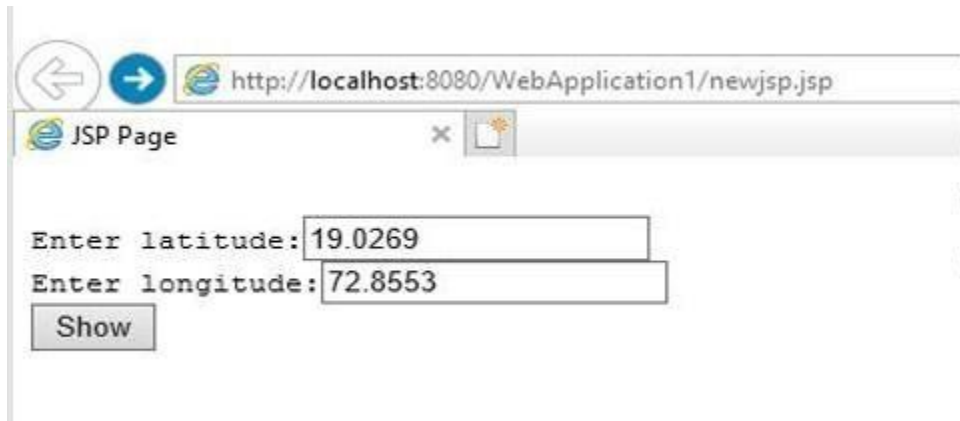
```
<style>
#map{
height: 400px;
width: 100%;
}
</style>
</head>
<body>
<%
double lati=Double.parseDouble(request.getParameter("t1"));
double longi=Double.parseDouble(request.getParameter("t2"));
%>
<h3> Google Maps</h3>
<div id="map"></div>
<script lang="javascript">
function initMap() {
var info={lat: <%=lati%>, lng: <%=longi%>};
var map =newgoogle.maps.Map(document.getElementById('map'),
{
zoom: 4, center: info
});
var marker = new google.maps.Marker({
position: info,
map: map
});
}
</script>
<script async defer
src="put your key here">
```

</script>

</body>

</html>

Output: -



Practical 5

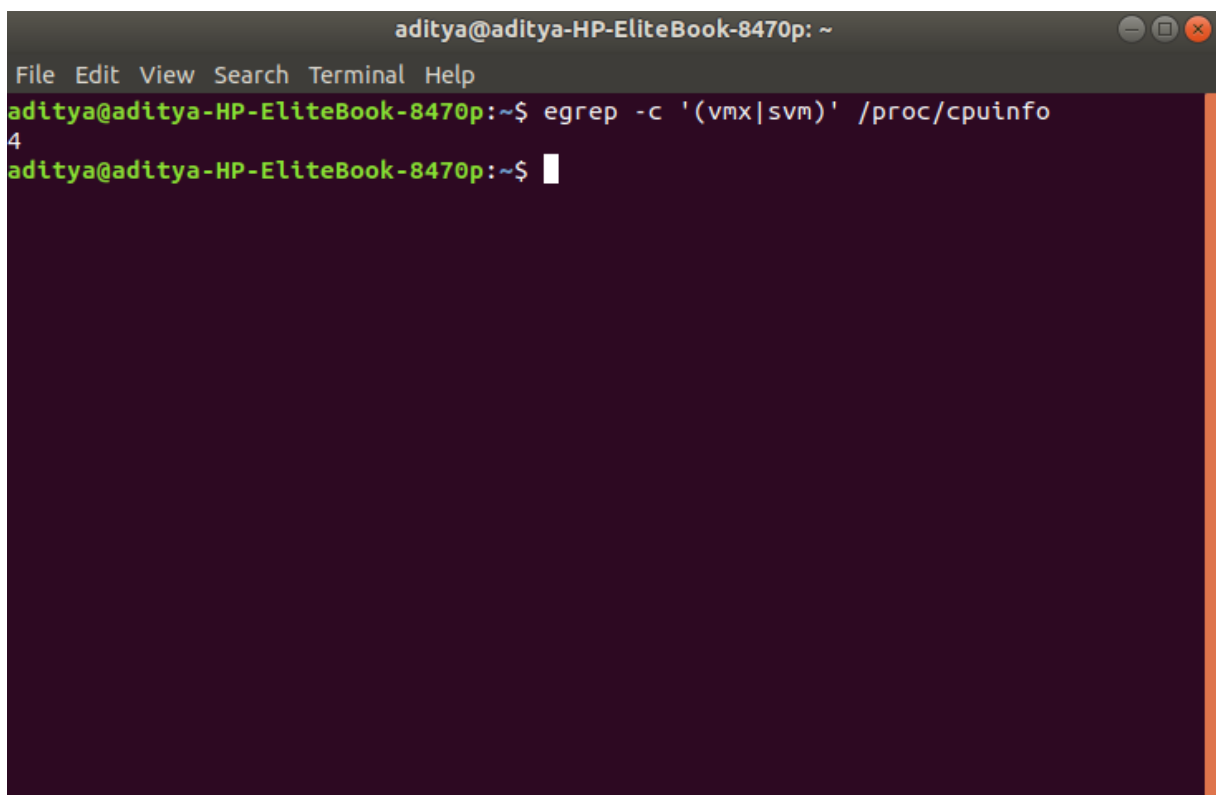
Aim: Installation and Configuration of virtualization using KVM.

Steps to install Virtualization using KVM

To create a virtual machine first we need to ensure that virtualization is enabled on our system. It is mandatory to create virtual machines. There are multiple ways to check if virtualization is enabled,

```
$ egrep -c '(vmx|svm)' /proc/cpuinfo
```

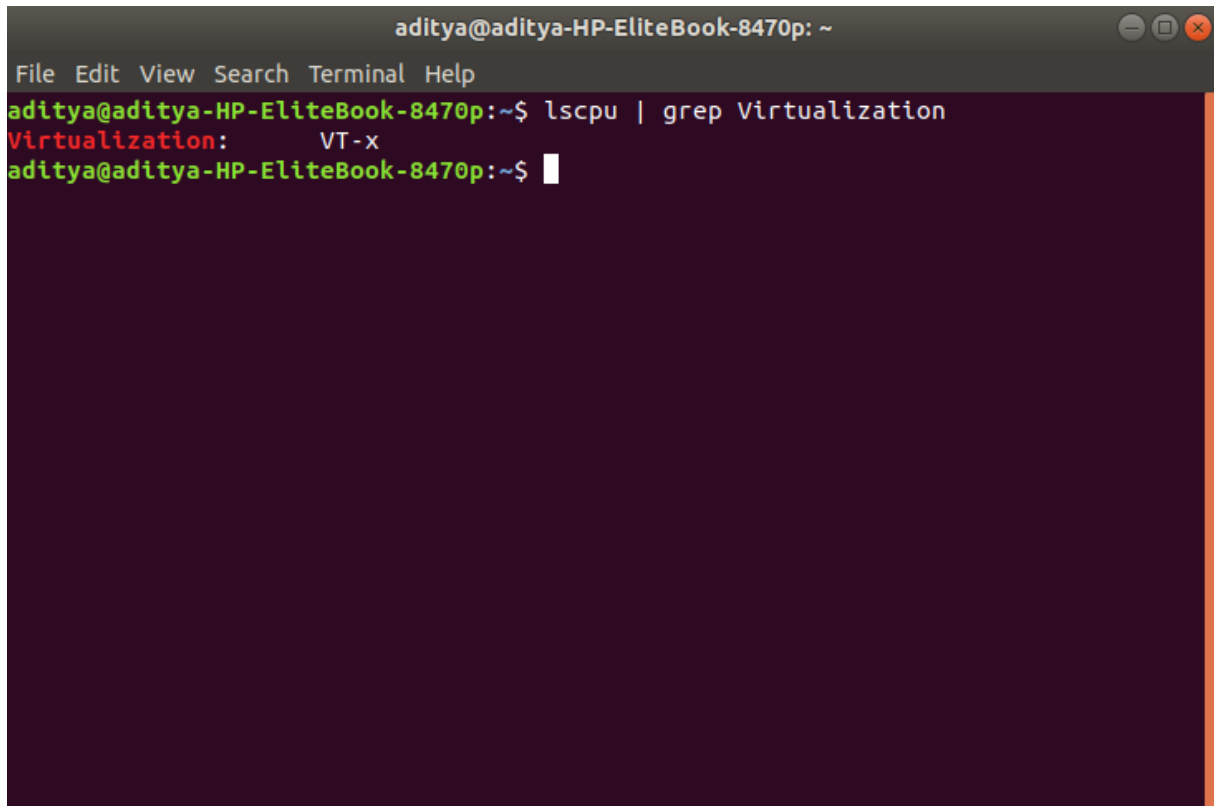
The above command `/proc/cpuinfo` gives information about the processor. The output of the command will be a number. The output number 1 or more than that represents that virtualization is enabled, output 0 says you need to enable virtualization on your system.

A screenshot of a terminal window titled 'aditya@aditya-HP-EliteBook-8470p: ~'. The terminal shows the command 'egrep -c '(vmx|svm)' /proc/cpuinfo' being executed, and the output '4' is displayed on the next line. The terminal has a dark purple background and a light blue prompt. The window has standard Linux window controls (minimize, maximize, close) in the top right corner.

Virtualization Inability

```
$ lscpu | grep Virtualization
```

This command is used to check which type of virtualization your processor supports. If the system contains a CPU with Intel VT support, the above command will provide the following output

A terminal window titled 'aditya@aditya-HP-EliteBook-8470p: ~' with a menu bar (File, Edit, View, Search, Terminal, Help). The prompt is 'aditya@aditya-HP-EliteBook-8470p:~\$'. The command 'lscpu | grep Virtualization' has been entered, and the output 'Virtualization: VT-x' is displayed. The prompt is now 'aditya@aditya-HP-EliteBook-8470p:~\$' with a cursor.

```
aditya@aditya-HP-EliteBook-8470p: ~
File Edit View Search Terminal Help
aditya@aditya-HP-EliteBook-8470p:~$ lscpu | grep Virtualization
Virtualization:      VT-x
aditya@aditya-HP-EliteBook-8470p:~$
```

Virtualization Type

Installing KVM on Ubuntu

`$ sudo apt-get install virt-manager`

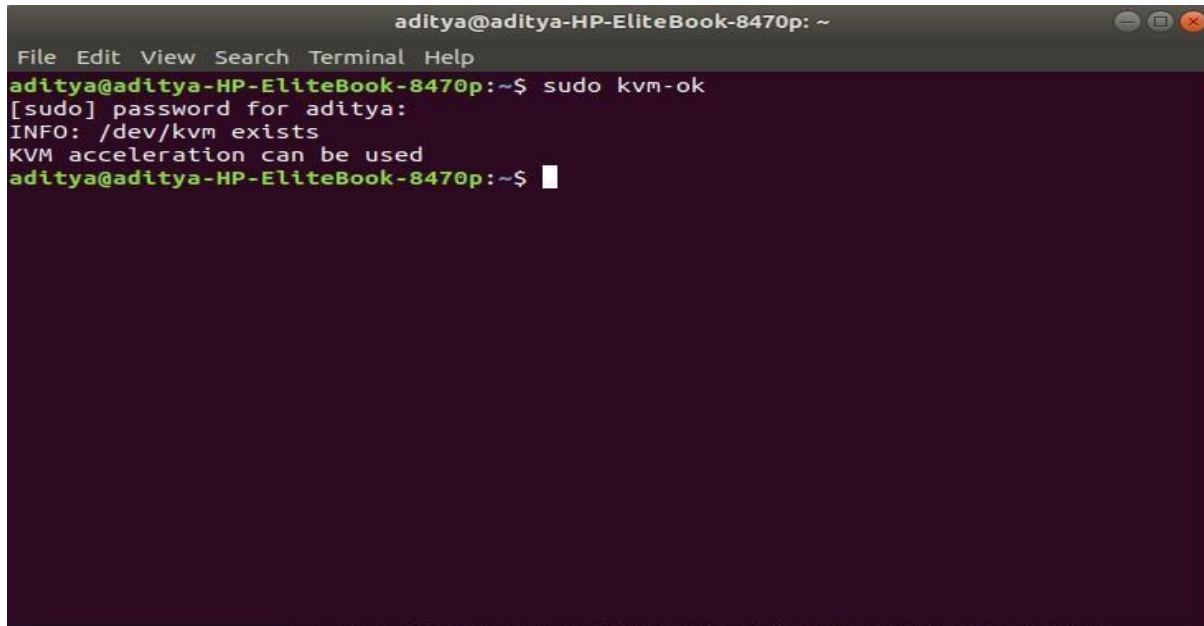
Now we know our system is capable of creating and running virtual machines, it's time to install tools which will create our virtual machines. To install KVM and other KVM dependencies such as virt-manager, bridge-utils enter command:

`$ sudo apt -y install bridge-utils cpu-checker libvirt-clients libvirt-daemon qemu qemu-kvm`

1. bridge-utils: The bridge-utils package contains a utility needed to create and manage bridge devices. This is useful in setting up networks for a hosted virtual machine (VM).
2. cpu-checker: Outputs the specifications of CPU
3. libvirt-clients: a toolkit to manage virtualization platforms/clients and hypervisors
4. qemu: A program that can run the operating system of the machine on different machines
5. qemu-kvm: Runs process using KVM module

All dependencies are installed now run command its time to check if KVM ins installed successfully:

\$ sudo kvm-ok

A terminal window titled 'aditya@aditya-HP-EliteBook-8470p: ~' with a menu bar (File, Edit, View, Search, Terminal, Help). The terminal shows the command 'sudo kvm-ok' being executed. The output is: '[sudo] password for aditya:', 'INFO: /dev/kvm exists', and 'KVM acceleration can be used'. The prompt returns to 'aditya@aditya-HP-EliteBook-8470p:~\$'.

Check if KVM is Installed properly

Also, we need to confirm if the virtualization daemon – libvirt-daemon – is running, to do so enter the command.

\$ sudo systemctl status libvirt

If the output is not active: running you need to start daemon thread

```

aditya@aditya-HP-EliteBook-8470p: ~
File Edit View Search Terminal Help
● libvirtd.service - Virtualization daemon
   Loaded: loaded (/lib/systemd/system/libvirtd.service; enabled; vendor preset:
   Active: active (running) since Tue 2022-02-22 23:00:23 IST; 1h 22min ago
     Docs: man:libvirtd(8)
           https://libvirt.org
   Main PID: 8406 (libvirtd)
     Tasks: 33 (limit: 32768)
    CGroup: /system.slice/libvirtd.service
            └─ 3784 /usr/sbin/dnsmasq --conf-file=/var/lib/libvirt/dnsmasq/default
            └─ 3785 /usr/sbin/dnsmasq --conf-file=/var/lib/libvirt/dnsmasq/default
            └─ 8406 /usr/sbin/libvirtd
            └─ 9405 qemu-system-x86_64 -enable-kvm -name guest=ubuntu-guest,debug
            └─ 11366 qemu-system-x86_64 -enable-kvm -name guest=ubuntu18.04,debug-

Feb 22 23:00:27 aditya-HP-EliteBook-8470p libvirtd[8406]: 2022-02-22 17:30:27.13
Feb 22 23:00:27 aditya-HP-EliteBook-8470p libvirtd[8406]: 2022-02-22 17:30:27.18
Feb 22 23:00:27 aditya-HP-EliteBook-8470p libvirtd[8406]: 2022-02-22 17:30:27.18
Feb 22 23:05:20 aditya-HP-EliteBook-8470p libvirtd[8406]: 2022-02-22 17:35:20.15
Feb 22 23:10:55 aditya-HP-EliteBook-8470p dnsmasq-dhcp[3784]: DHCPDISCOVER(virbr
Feb 22 23:10:55 aditya-HP-EliteBook-8470p dnsmasq-dhcp[3784]: DHCPOFFER(virbr0)
Feb 22 23:10:55 aditya-HP-EliteBook-8470p dnsmasq-dhcp[3784]: DHCPDISCOVER(virbr
Feb 22 23:10:55 aditya-HP-EliteBook-8470p dnsmasq-dhcp[3784]: DHCPOFFER(virbr0)
Feb 22 23:10:55 aditya-HP-EliteBook-8470p dnsmasq-dhcp[3784]: DHCPREQUEST(virbr0
lines 1-23

```

If the daemon thread is not running enter the following command to start the thread,

```
$ sudo systemctl enable --now libvirtd
```

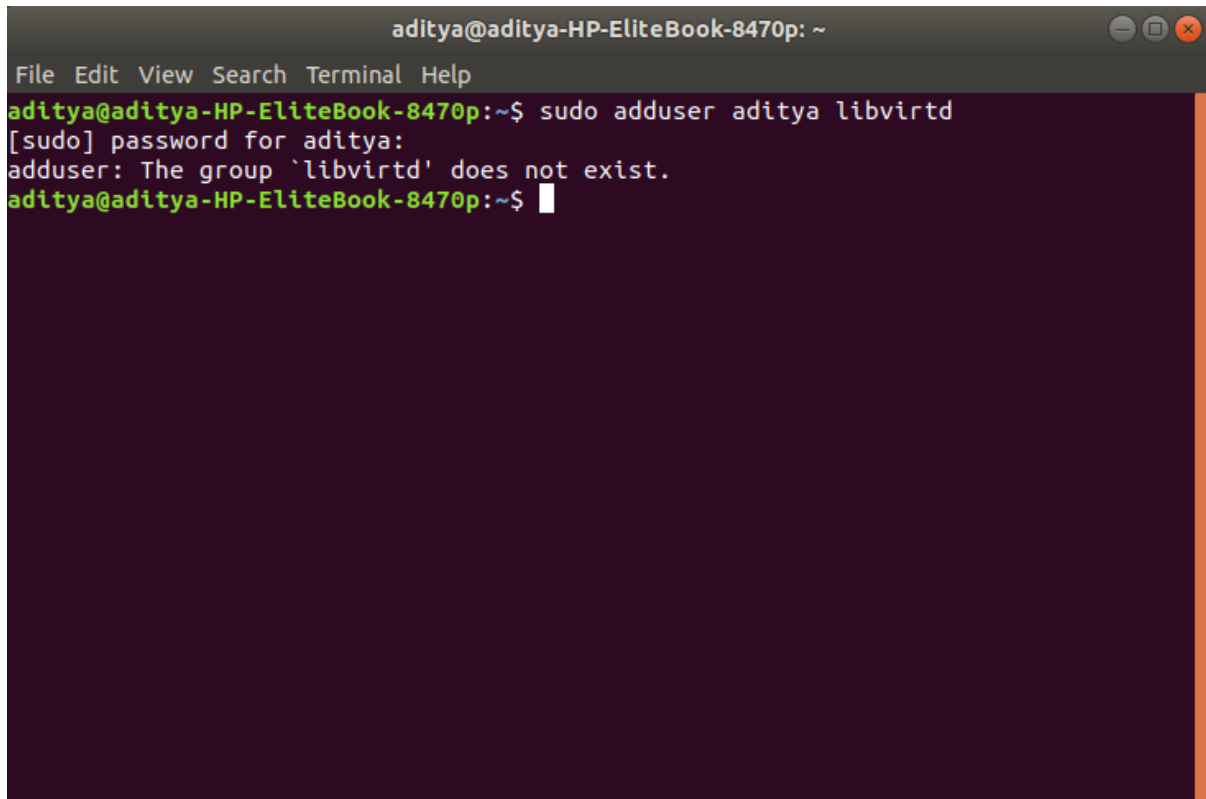
Adding a user to KVM

In this part, we are going to create a user for KVM. To prevent root user from using KVM and root user is only available when root user is a part of libvirt/libvirtd group.

To add a new user to KVM, use the following command,

```
sudo adduser [username] libvirtd
```

[username] enter the username of your choice if the output is

A terminal window titled 'aditya@aditya-HP-EliteBook-8470p: ~' with a menu bar (File, Edit, View, Search, Terminal, Help). The terminal shows the command 'aditya@aditya-HP-EliteBook-8470p:~\$ sudo adduser aditya libvirt'. It prompts for a password, then shows the error 'adduser: The group `libvirt' does not exist.' followed by the prompt 'aditya@aditya-HP-EliteBook-8470p:~\$'.

Adding User to KVM

If your KVM is already a member of the non-root user and serves the same purpose as libvirt group then you don't need to add yourself to the group.

Creating Virtual Machine

There are two ways to create a virtual machine

1. Using the command line
 2. Using the graphical interface
- **Create a Virtual Machine via Command Line**

virt-install is a command which is used to create virtual machines in Linux, following is the command which creates a VM.

```
sudo virt-install --name=ubuntu-guest --os-variant=ubuntu20.04 --vcpu=2 --ram=2048 --  
graphics none --location=[local path to iso file] --network bridge:vibr0
```

The above command creates a Ubuntu virtual machine with version 20.04 and the name ubuntu-guest.

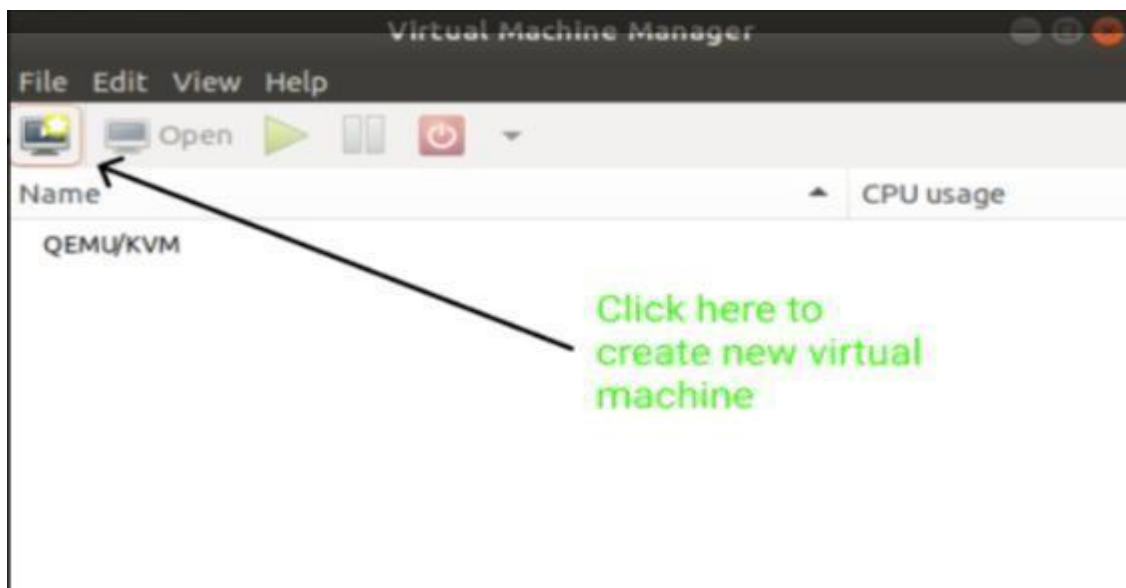
1. Name: specify the name of the virtual machine being created
2. vcpu: Number of virtual CPUs to configure for the guest.

3. Ram: Memory to allocate for guest instance in megabytes. According to your machine, you can specify the given memory of the VM.
4. Graphics spice: If no graphics option is specified, “virt-install” will default to ‘–graphics vnc’ if the display environment variable is set, otherwise ‘–graphics none’ is used.
5. Location: location of iso file on which virtual machine will be built. It can be the path to an ISO image, or to a CDROM device. It can also be a URL from which to fetch/access a minimal boot ISO image.
6. Network-bridge: Connect the guest to the host network, Connect to a bridge device in the host called “BRIDGE”.

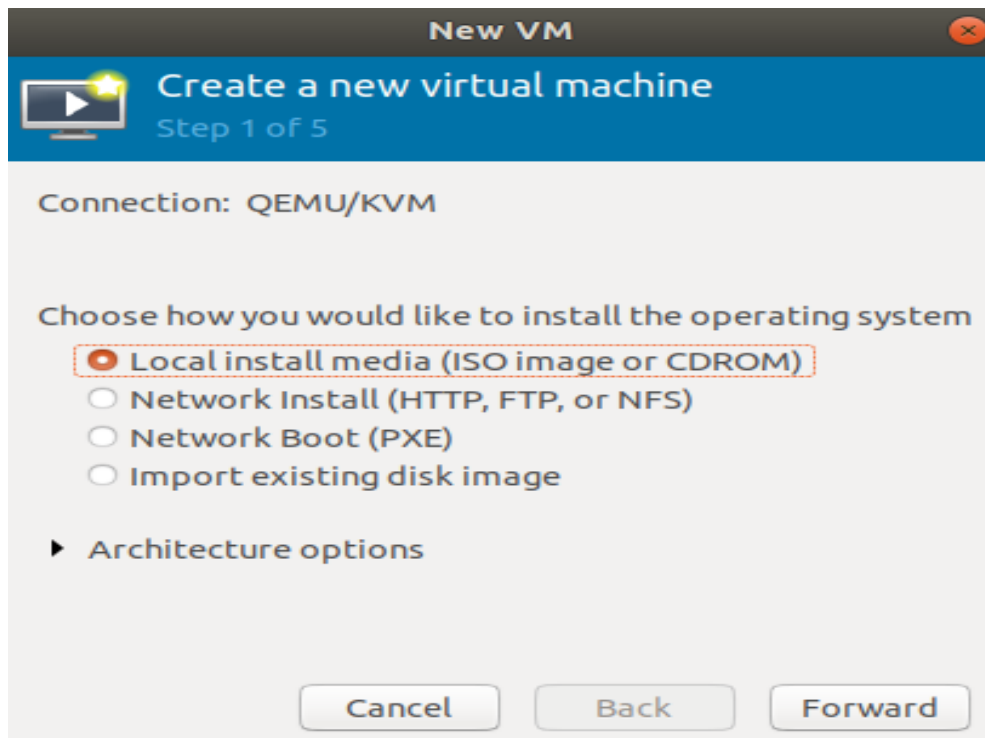
Create a Virtual machine using a graphical interface

If you are not much familiar with the command line don't worry there's another way to create a virtual machine using a tool called virt-manager you can easily create virtual machines. Steps to create VM using virt-manager,

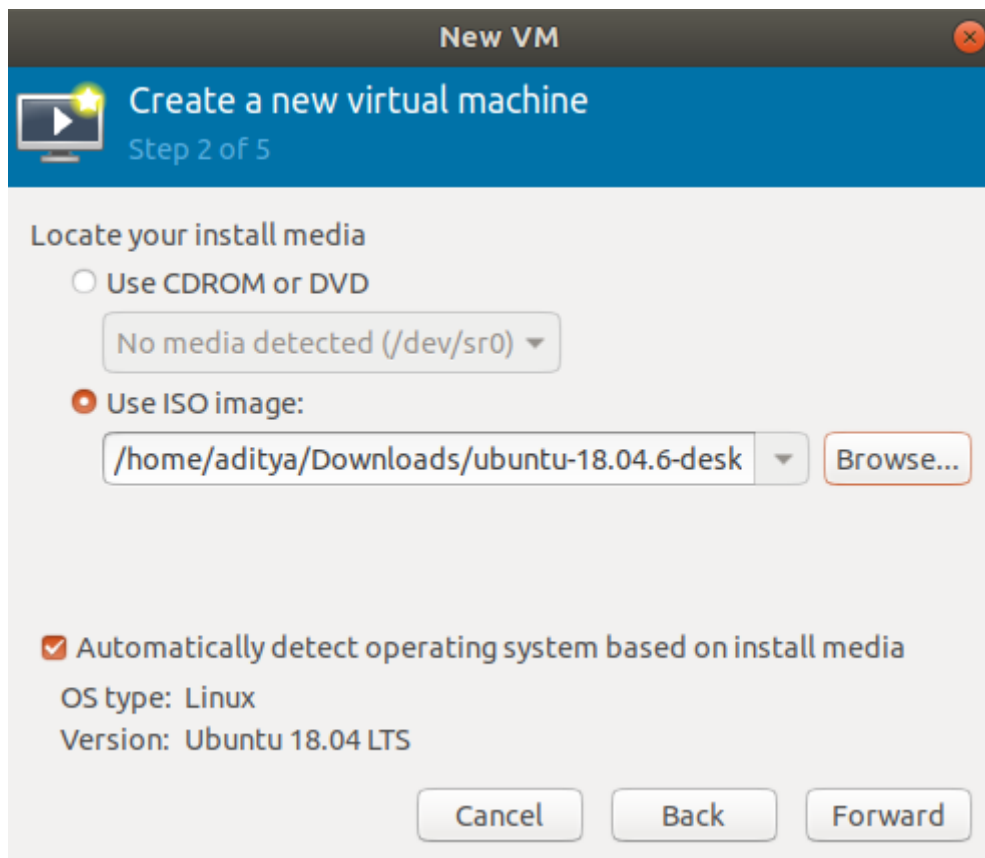
Step 1: Launch virt-manager



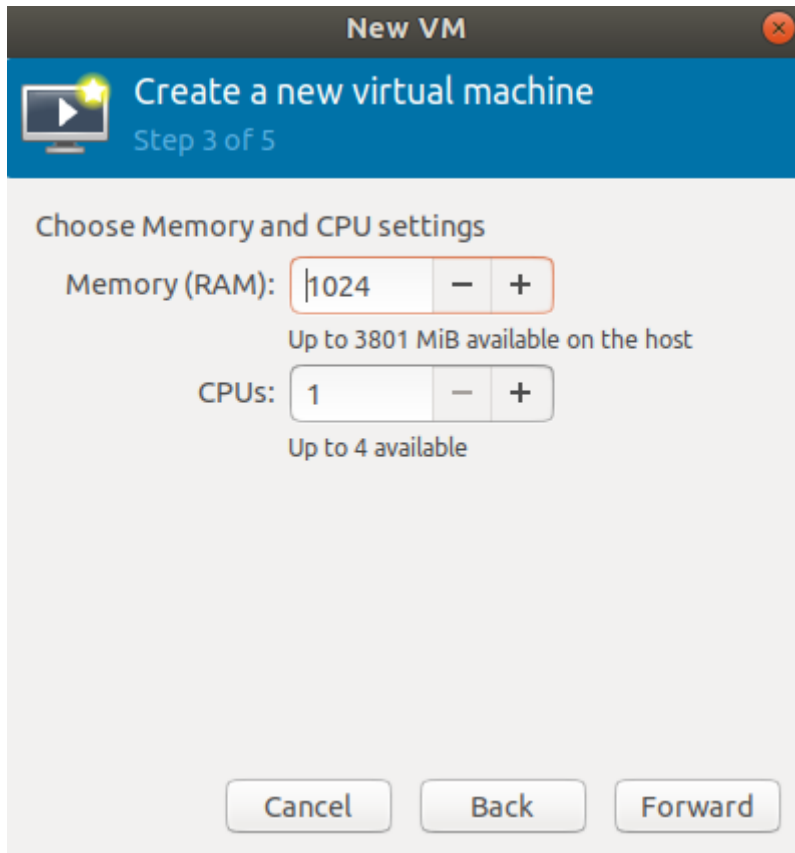
Step 2: Choose installation media it can be an iso file of OS, you can install from the network or can be a disk copy



Step 3: Enter the path/URL of your file click next



Step 4: Enter memory and CPU requirements



New VM

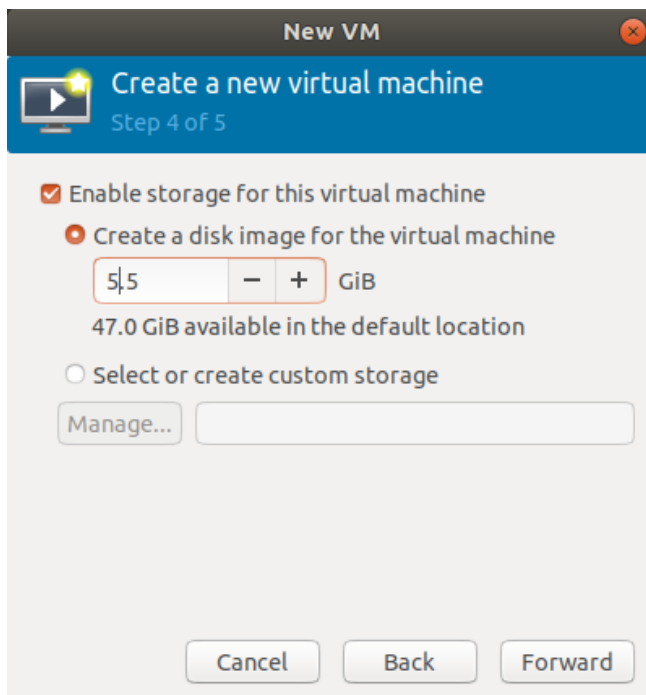
Create a new virtual machine
Step 3 of 5

Choose Memory and CPU settings

Memory (RAM): 1024 - +
Up to 3801 MiB available on the host

CPUs: 1 - +
Up to 4 available

Cancel Back Forward

Step 5: Enter required disk space

New VM

Create a new virtual machine
Step 4 of 5

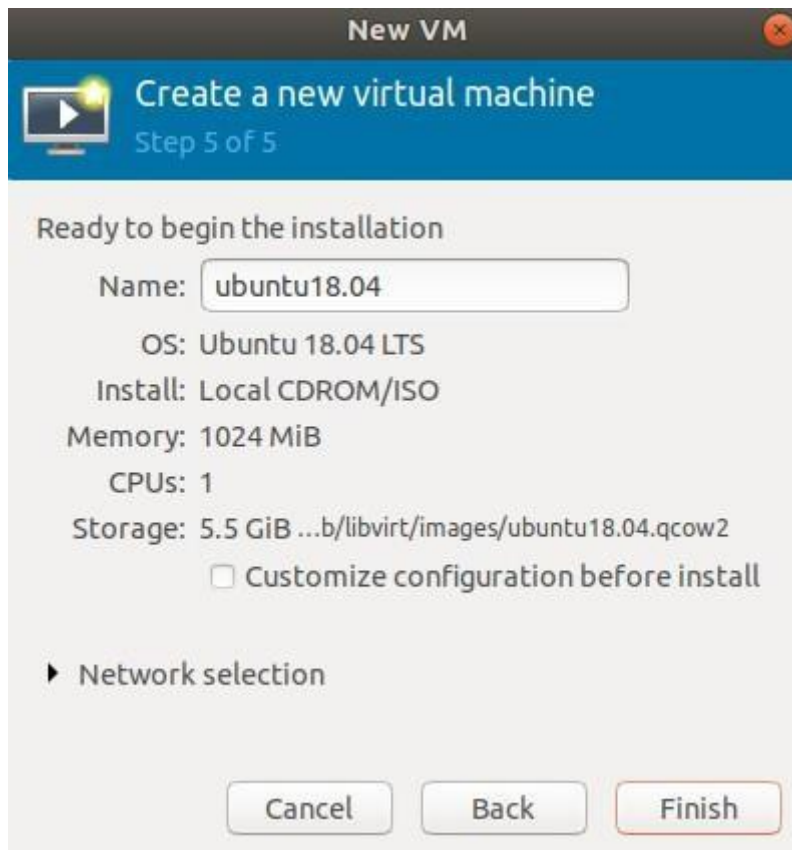
☒ Enable storage for this virtual machine

☒ Create a disk image for the virtual machine
5.5 - + GiB
47.0 GiB available in the default location

☐ Select or create custom storage
Manage...

Cancel Back Forward

Step 6: Enter the name of VM and check specifications and hit the finish



Now the installations should be completed and the virtual machine should start running.

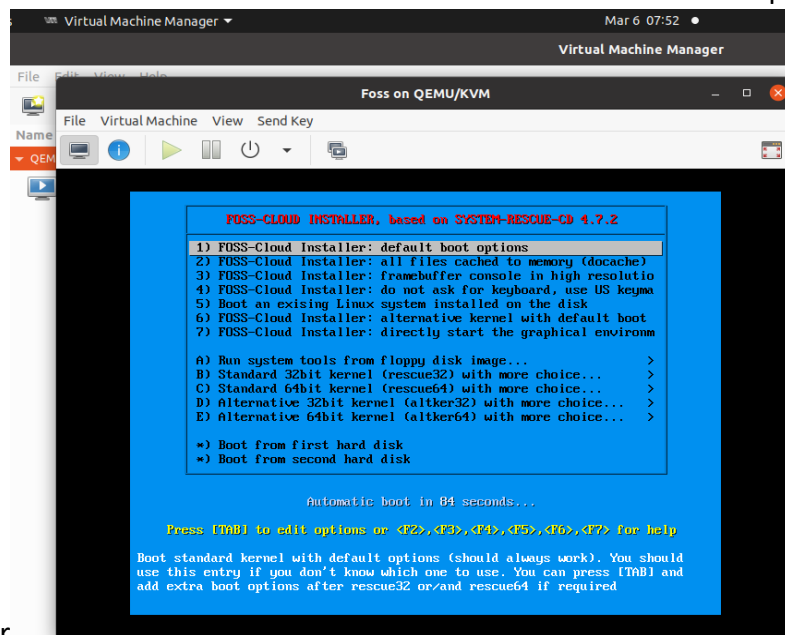
Conclusion: Hence we have successfully installed and Configured virtualization using KVM.

Practical 6

AIM: Implement FOSS-Cloud Functionality VSI (Virtual Server Infrastructure) Infrastructure as a Service (IaaS), Storage.

Steps:

1. Start VMWARE workstation player 15 ? select Open virtual machine then? Browse a foss cloud iso file? then select Operating System: Other Version: Other, Name: FOSS.
2. Then click on edit virtual machine and select Processor: 2 or 3, Memory: expand to 165 GB.
3. Once the foss Cloud is launched select “Foss-Cloud installer :default boot options”

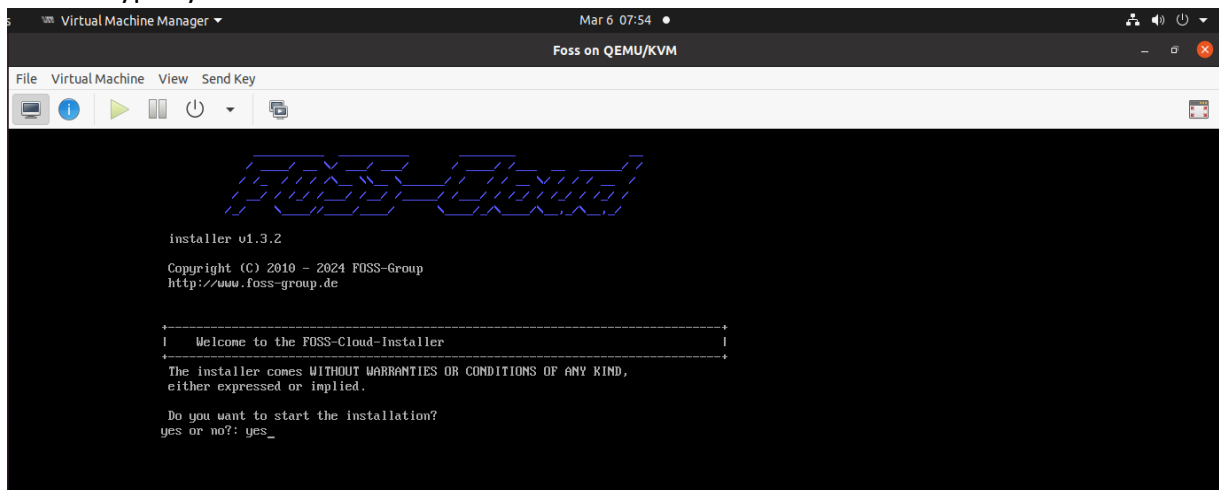


and press Enter

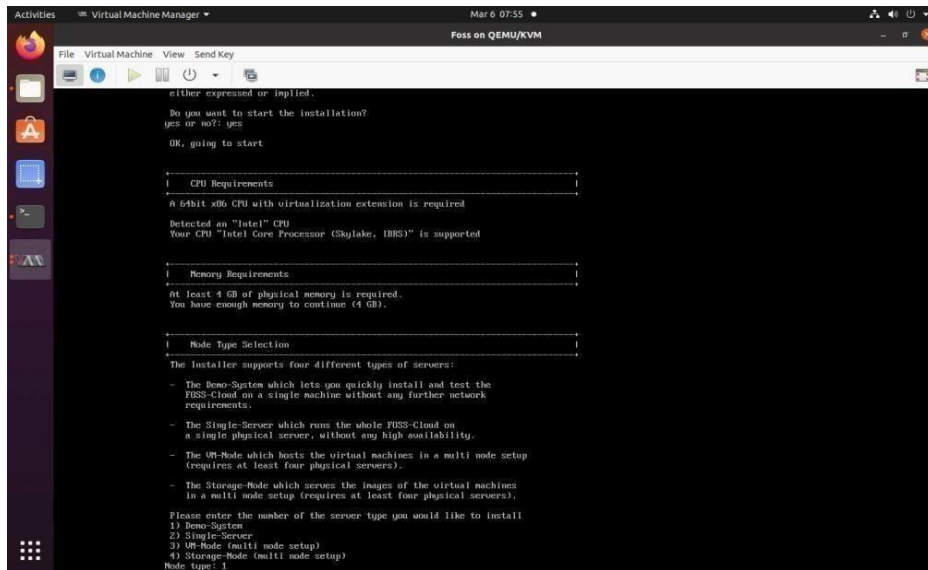
4. Reboot your system and select foss cloud



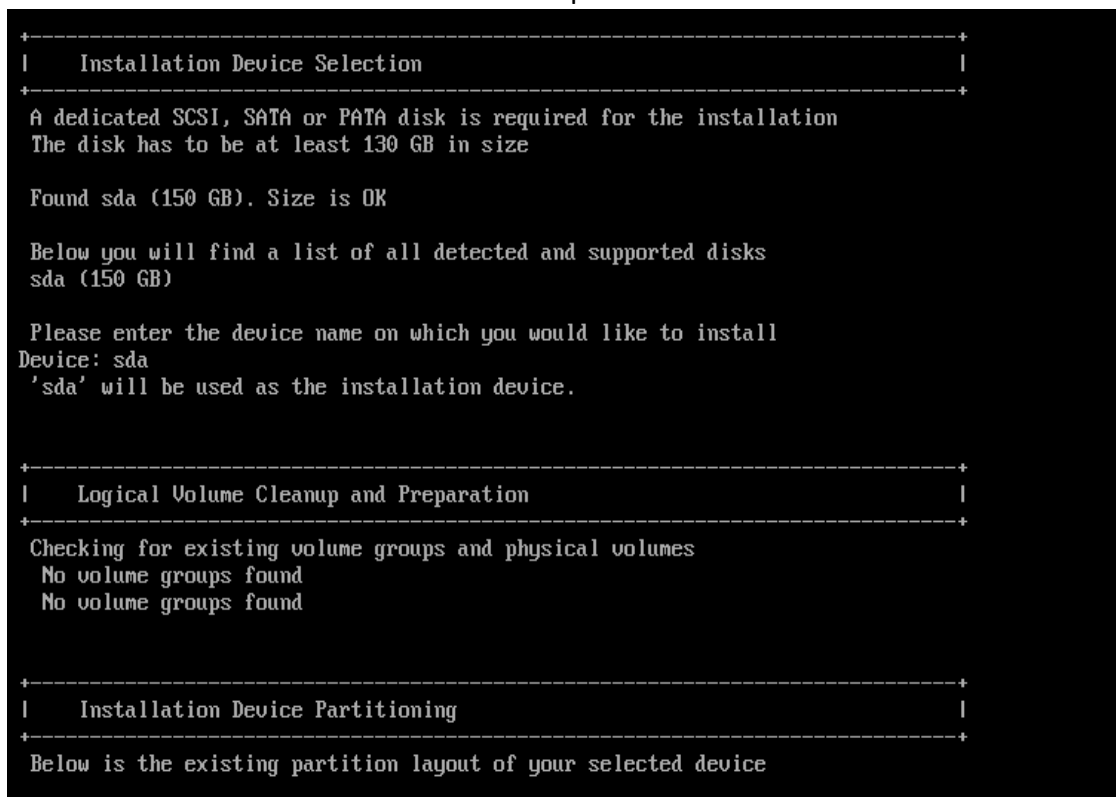
5. Types yes to start the installation



6. Select "Demo System" and press Enter



7. Enter the Device name as “sda” and press enter



8. Put credential of foss cloud and start foss cloud

Virtual Machine Manager

Mar 6 08:02

Foss on QEMU/KVM

File Virtual Machine View Send Key

```
+-----+
| Logical Volume Setup |
+-----+

Setup LVM environment and volumes
No volume groups found
Volume group "local0" successfully created
Logical volume "home" created
Logical volume "var" created
Logical volume "tmp" created
Logical volume "portage" created
Logical volume "virtualization" created

All LVM volumes created successfully

+-----+
| Filesystem Creation |
+-----+

Creating filesystems
mke2fs 1.42.13 (17-May-2015)

Filesystem creation was successful

+-----+
| Mounting Filesystems |
+-----+

Mounting of filesystems was successful

+-----+
| Stage4 Installation |
+-----+

Unpacking stage4 tarball
This will take a while - please be patient

Unpacking of stage4 tarball was successful

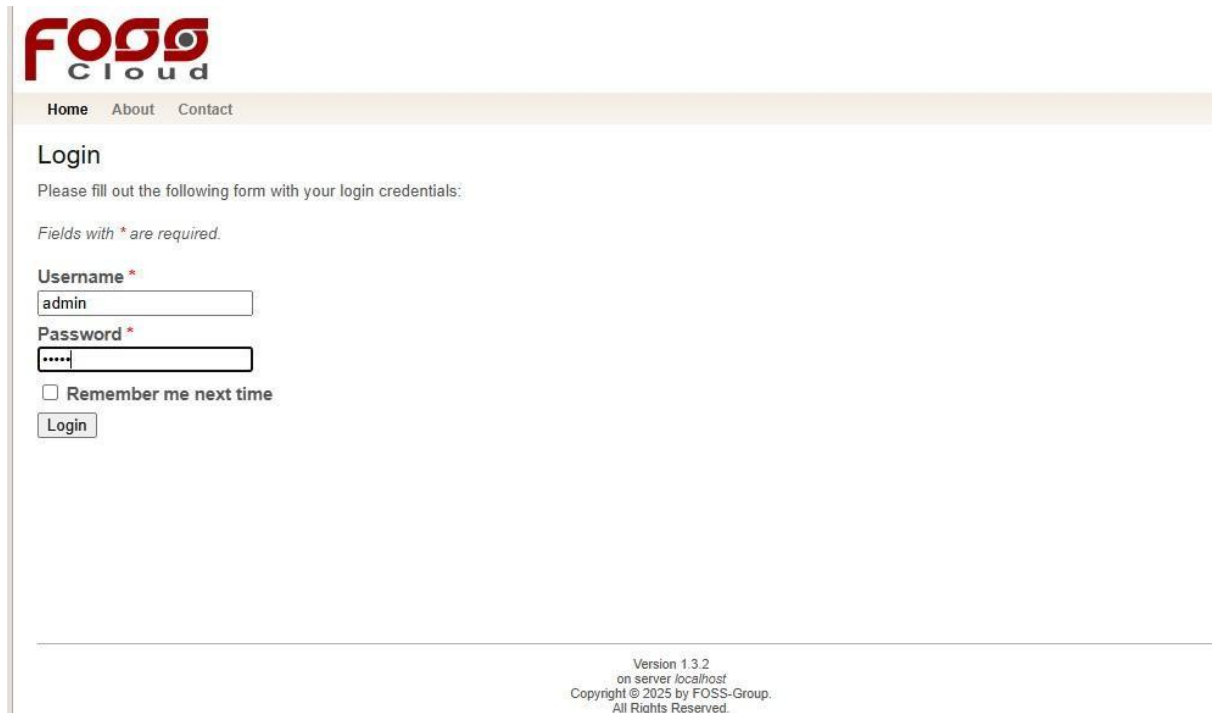
+-----+
| Network Device Selection |
+-----+

Please enter the device which you would like to use

Available ethernet devices: ens3
Device #0:
```

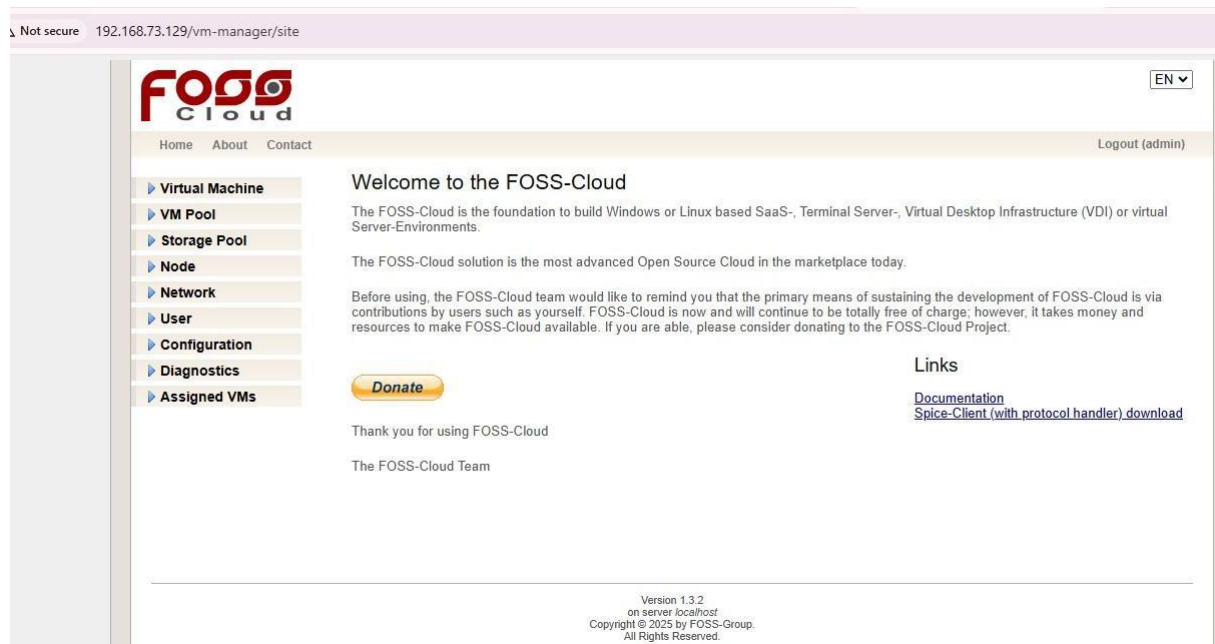
[illegible]

9. Then Execute below mention command



The image shows the login page of the FOSS Cloud interface. At the top left is the FOSS Cloud logo. Below it is a navigation bar with links for Home, About, and Contact. The main heading is 'Login'. Below the heading, it says 'Please fill out the following form with your login credentials:'. A note states 'Fields with * are required.' The form includes a 'Username *' field with the value 'admin', a 'Password *' field with masked characters '****', a checkbox for 'Remember me next time', and a 'Login' button. At the bottom right, it says 'Version 1.3.2 on server localhost Copyright © 2025 by FOSS-Group. All Rights Reserved.'

12. After putting the login details you will get the below mention page



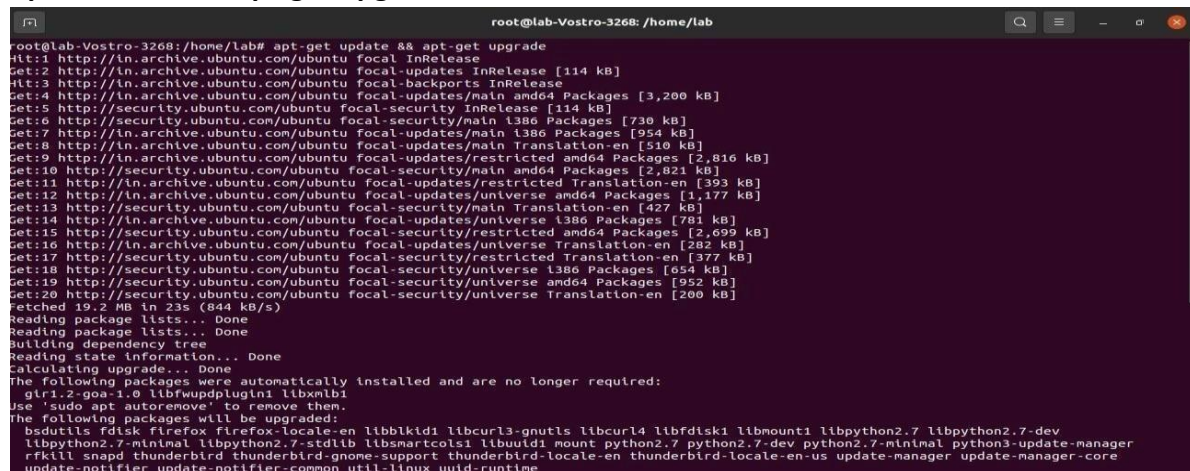
The image shows the dashboard of the FOSS Cloud interface after a successful login. The top bar shows 'Not secure 192.168.73.129/vm-manager/site' and a language dropdown set to 'EN'. The dashboard has a sidebar with a list of menu items: Virtual Machine, VM Pool, Storage Pool, Node, Network, User, Configuration, Diagnostics, and Assigned VMs. The main content area has a heading 'Welcome to the FOSS-Cloud' and a paragraph explaining the platform. Below this is a 'Donate' button and a 'Thank you for using FOSS-Cloud' message from 'The FOSS-Cloud Team'. On the right, there is a 'Logout (admin)' link and a 'Links' section with links to 'Documentation' and 'Spice-Client (with protocol handler) download'. At the bottom right, it says 'Version 1.3.2 on server localhost Copyright © 2025 by FOSS-Group. All Rights Reserved.'

Conclusion: Hence, we have successfully installed FOSS-Cloud Functionality VSI (Virtual Server Infrastructure) Infrastructure as a Service (IaaS), Storage.

Practical 7

AIM: Using AWS Flow Framework develop applications that include a simple workflow. Workflow calls an activity to print hello world to the console. It must define the basic usage of AWS Flow Framework, including defining contracts, implementation of activities and workflow coordination logic and worker programs to host them.

Step 1: Open Terminal and Update and Upgrade your system by command `sudo apt-get update && sudo apt-get upgrade`



```

root@lab-Vostro-3268: /home/lab# sudo apt-get update && apt-get upgrade
Hit:1 http://in.archive.ubuntu.com/ubuntu focal InRelease
Get:2 http://in.archive.ubuntu.com/ubuntu focal-updates InRelease [114 kB]
Hit:3 http://in.archive.ubuntu.com/ubuntu focal-backports InRelease
Get:4 http://in.archive.ubuntu.com/ubuntu focal-updates/main amd64 Packages [3,200 kB]
Get:5 http://security.ubuntu.com/ubuntu focal-security InRelease [114 kB]
Get:6 http://security.ubuntu.com/ubuntu focal-security/main i386 Packages [730 kB]
Get:7 http://in.archive.ubuntu.com/ubuntu focal-updates/main i386 Packages [954 kB]
Get:8 http://in.archive.ubuntu.com/ubuntu focal-updates/main Translation-en [510 kB]
Get:9 http://in.archive.ubuntu.com/ubuntu focal-updates/restricted amd64 Packages [2,816 kB]
Get:10 http://security.ubuntu.com/ubuntu focal-security/main amd64 Packages [2,821 kB]
Get:11 http://in.archive.ubuntu.com/ubuntu focal-updates/restricted Translation-en [393 kB]
Get:12 http://in.archive.ubuntu.com/ubuntu focal-updates/universe amd64 Packages [1,177 kB]
Get:13 http://security.ubuntu.com/ubuntu focal-security/main Translation-en [427 kB]
Get:14 http://in.archive.ubuntu.com/ubuntu focal-updates/universe i386 Packages [781 kB]
Get:15 http://security.ubuntu.com/ubuntu focal-security/restricted amd64 Packages [2,699 kB]
Get:16 http://in.archive.ubuntu.com/ubuntu focal-updates/universe Translation-en [282 kB]
Get:17 http://security.ubuntu.com/ubuntu focal-security/restricted Translation-en [377 kB]
Get:18 http://security.ubuntu.com/ubuntu focal-security/universe i386 Packages [854 kB]
Get:19 http://security.ubuntu.com/ubuntu focal-security/universe amd64 Packages [952 kB]
Get:20 http://security.ubuntu.com/ubuntu focal-security/universe Translation-en [200 kB]
Fetched 19.2 MB in 23s (844 kB/s)
Reading package lists... Done
Reading package lists... Done
Building dependency tree
Reading state information... Done
Calculating upgrade... Done
The following packages were automatically installed and are no longer required:
  glibc-gconv-locales libfwupdplugin1 libxmb1
Use 'sudo apt autoremove' to remove them.
The following packages will be upgraded:
  bsdutils fdisk firefox firefox-locale-en libblkid1 libcurl3-gnutls libcurl4 libfdisk1 libmount1 libpython2.7 libpython2.7-dev
  libpython2.7-minimal libpython2.7-stdlib libsmartcols1 libuuid1 mount python2.7 python2.7-dev python2.7-minimal python3-update-manager
  rfskill snapd thunderbird thunderbird-gnome-support thunderbird-locale-en thunderbird-locale-en-us update-manager update-manager-core
  update-notifier update-notifier-common util-linux uuid-runtime
100% [100] 57.5M [100] 57.5M [0] 0 2866k [0] 0:00:20 [0:00:20] --:--:-- 4225k
root@lab-Vostro-3268: /home/lab#

```

Step 2: Download awscli2.zip with command

`curl "https://awscli.amazonaws.com/awscli-exe-linux-x86_64.zip" -o "awscli2.zip"`



```

root@lab-Vostro-3268: /home/lab# curl "https://awscli.amazonaws.com/awscli-exe-linux-x86_64.zip" -o "awscli2.zip"
% Total    % Received % Xferd Average Speed   Time    Time     Time  Current
           Dload  Upload   Total   Spent    Left  Speed
100 57.5M  100 57.5M    0     0 2866k      0  0:00:20  0:00:20 --:--:-- 4225k
root@lab-Vostro-3268: /home/lab#

```

Step 3: Download awscli2.sig file with command

`curl -o awscli2.sig https://awscli.amazonaws.com/awscli-exe-linux-x86_64.zip.sig`

```

root@lab-Vostro-3268: /home/lab

root@lab-Vostro-3268:/home/lab# curl -o awscliv2.sig https://awscli.amazonaws.com/awscli-exe-linux-x86_64.zip.sig
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload  Total  Spent    Left  Speed
100 566 100 566    0    0  765      0 --:--:-- --:--:-- --:--:-- 765
root@lab-Vostro-3268:/home/lab#

```

Step4: Verify installation

`sam --version`

Step 5: Type command pip3 install aws-sam-cli

```

root@lab-Vostro-3268:/home/lab# pip3 install aws-sam-cli
Collecting aws-sam-cli
  Downloading aws_sam_cli-1.113.0-py3-none-any.whl (5.9 MB)
    | 5.9 MB 21 kB/s
Collecting aws-lambda-builders==1.47.0
  Downloading aws_lambda_builders-1.47.0-py3-none-any.whl (130 kB)
    | 130 kB 547 kB/s
Collecting tzlocal==5.2
  Downloading tzlocal-5.2-py3-none-any.whl (17 kB)
Collecting dateparser==1.2
  Downloading dateparser-1.2.0-py2.py3-none-any.whl (294 kB)
    | 294 kB 2.4 MB/s
Collecting Flask<3.1
  Downloading flask-3.0.2-py3-none-any.whl (101 kB)
    | 101 kB 447 kB/s
Collecting boto3<2,>=1.29.2
  Downloading boto3-1.34.76-py3-none-any.whl (139 kB)
    | 139 kB 558 kB/s
Collecting pyopenssl~=24.1.0
  Downloading pyOpenSSL-24.1.0-py3-none-any.whl (56 kB)
    | 56 kB 580 kB/s
Collecting requests~=2.31.0
  Using cached requests-2.31.0-py3-none-any.whl (62 kB)
Collecting click~=8.1
  Downloading click-8.1.7-py3-none-any.whl (97 kB)
    | 97 kB 377 kB/s
Collecting watchdog==4.0.0
  Downloading watchdog-4.0.0-py3-none-manylinux2014_x86_64.whl (82 kB)
    | 82 kB 113 kB/s
Collecting chevron~=0.12
  Downloading chevron-0.14.0-py3-none-any.whl (11 kB)
Collecting ruamel-yaml~=0.18.6
  Downloading ruamel.yaml-0.18.6-py3-none-any.whl (117 kB)
    | 117 kB 363 kB/s
Collecting aws-sam-translator==1.86.0

```

Step 6: Type command sam init in terminal to launch Sam

```

root@lab-Vostro-3268:/home/lab# sam init

SAM CLI now collects telemetry to better understand customer needs.

You can OPT OUT and disable telemetry collection by setting the
environment variable SAM_CLI_TELEMETRY=0 in your shell.
Thanks for your help!

Learn More: https://docs.aws.amazon.com/serverless-application-model/latest/developerguide/serverless-sam-telemetry.html

/usr/lib/python3/dist-packages/paramiko/transport.py:220: CryptographyDeprecationWarning: Blowfish has been deprecated and will be removed in
a future release
  "class": algorithms.Blowfish,

You can preselect a particular runtime or package type when using the 'sam init' experience.
Call 'sam init --help' to learn more.

Which template source would you like to use?
  1 - AWS Quick Start Templates
  2 - Custom Template Location
Choice: 1

```

CLI Select 1st option to use AWS Quick Start Templates**Step 7: Select Template no.1 Hello World Example**

```
Choose an AWS Quick Start application template
 1 - Hello World Example
 2 - Data processing
 3 - Hello World Example with Powertools for AWS Lambda
 4 - Multi-step workflow
 5 - Scheduled task
 6 - Standalone function
 7 - Serverless API
 8 - Infrastructure event management
 9 - Lambda Response Streaming
10 - Serverless Connector Hello World Example
11 - Multi-step workflow with Connectors
12 - GraphQLApi Hello World Example
13 - Full Stack
14 - Lambda EFS example
15 - Hello World Example With Powertools for AWS Lambda
16 - DynamoDB Example
17 - Machine Learning
Template: 1
```

Step 8: Type "N" if it ask to use most popular runtime and package type

Open new terminal by pressing ctrl+shift+T and check for python version by command
python --version

Select the option according to your python version in my case its option 19- python 3.11

```
Use the most popular runtime and package type? (Python and zip) [y/N]: n
Which runtime would you like to use?
 1 - aot.dotnet7 (provided.al2)
 2 - dotnet8
 3 - dotnet6
 4 - go1.x
 5 - go (provided.al2)
 6 - go (provided.al2023)
 7 - graalvm.java11 (provided.al2)
 8 - graalvm.java17 (provided.al2)
 9 - java21
10 - java17
11 - java11
12 - java8.al2
13 - nodejs20.x
14 - nodejs18.x
15 - nodejs16.x
16 - python3.9
17 - python3.8
18 - python3.12
19 - python3.11
20 - python3.10
21 - ruby3.2
22 - rust (provided.al2)
23 - rust (provided.al2023)
Runtime: 17
```


Step 9: Select package type as Zip

```
What package type would you like to use?
  1 - Zip
  2 - Image
Package type: 1
```

Step 10: Now choose Yes option everytime it ask .

Give project name as per your preference in my case its sam-app-test

```
Based on your selections, the only dependency manager available is pip.
We will proceed copying the template using pip.

Would you like to enable X-Ray tracing on the function(s) in your application? [y/N]: y
X-Ray will incur an additional cost. View https://aws.amazon.com/xray/pricing/ for more details

Would you like to enable monitoring using CloudWatch Application Insights?
For more info, please view https://docs.aws.amazon.com/AmazonCloudWatch/latest/monitoring/cloudwatch-application-insights.html [y/N]: y
AppInsights monitoring may incur additional cost. View https://docs.aws.amazon.com/AmazonCloudWatch/latest/monitoring/appinsights-what-is.html
AppInsights-pricing for more details

Would you like to set Structured Logging in JSON format on your Lambda functions? [y/N]: y
Structured Logging in JSON format might incur an additional cost. View https://docs.aws.amazon.com/lambda/latest/dg/monitoring-cloudwatchlogs.html#monitoring-cloudwatchlogs-pricing for more details

Project name [sam-app]: sam-app-test

Cloning from https://github.com/aws/aws-sam-cli-app-templates (process may take a moment)

-----
Generating application:
-----
Name: sam-app-test
Runtime: python3.8
Architectures: x86_64
Dependency Manager: pip
Application Template: hello-world
Output Directory: .
Configuration file: sam-app-test/samconfig.toml

Next steps can be found in the README file at sam-app-test/README.md
```

Step 11: Now one folder will be created by your provided project name go into that folder by command `cd (folder startname)`

After entering the project folder we will invoke the HelloWorldFunction by using command `sam local invoke "HelloWorldFunction"`

```
root@lab-Vostro-3268:/home/lab/sam-app-test# sam local invoke 'HelloWorldFunction'
/usr/lib/python3/dist-packages/paramiko/transport.py:220: CryptographyDeprecationWarning: Blowfish has been deprecated and will be removed in a future release
  "class": algorithms.Blowfish,
Invoking app.lambda_handler (python3.8)
Local image was not found.
Removing rapid images for repo public.ecr.aws/sam/emulation-python3.8
Building image.....
.....
Using local image: public.ecr.aws/lambda/python:3.8-rapid-x86_64.

Mounting /home/lab/sam-app-test/hello-world as /var/task:ro,delegated, inside runtime container
START RequestId: 065fefd3-5d41-4b87-bb31-8d820fb635e6 Version: $LATEST
END RequestId: 4b9baa5c-2523-443d-b340-f86e2b139f6a
REPORT RequestId: 4b9baa5c-2523-443d-b340-f86e2b139f6a Init Duration: 0.06 ms Duration: 99.11 ms Billed Duration: 100 ms Memory Size: 128 MB Max Memory Used: 128 MB
{"statusCode": 200, "body": "{\"message\": \"hello world\"}"}
root@lab-Vostro-3268:/home/lab/sam-app-test#
```

It should give you **Status Code: 200**

Use command `sudo snap install docker` if docker error occurs.

Command to install docker

1. `sudo apt update`
2. `sudo apt install -y docker.io`
3. `sudo systemctl start docker`
4. `sudo systemctl enable docker`
5. `docker --version`

OPTIONAL STEPS IF STILL YOU ARE GETTING ERROR

- `sudo usermod -aG docker $USER`
- `newgrp docker`
- `docker ps`

Step 13: Type command `sam local start-api` this will give you URL open it in any browser.

```
root@lab-Vostro-3268:/home/lab/sam-app-test# sam local start-api
/usr/lib/python3/dist-packages/paramiko/transport.py:220: CryptographyDeprecationWarning: Blowfish has been deprecated and will be removed in a future release
  "class": algorithms.Blowfish,
Initializing the lambda functions containers.
Local image was not found.
Removing rapid images for repo public.ecr.aws/sam/emulation-python3.11
Building image.....
Using local image: public.ecr.aws/lambda/python:3.11-rapid-x86_64.

Mounting /home/lab/sam-app-test/hello_world as /var/task:ro,delegated, inside runtime container
Containers Initialization is done.
Mounting HelloWorldFunction at http://127.0.0.1:3000/hello [GET]
You can now browse to the above endpoints to invoke your functions. You do not need to restart/reload SAM CLI while working on your functions,
changes will be reflected instantly/automatically. If you used sam build before running local commands, you will need to re-run sam build for
the changes to be picked up. You only need to restart SAM CLI if you update your AWS SAM template
2024-04-03 09:26:18 WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on http://127.0.0.1:3000
2024-04-03 09:26:18 Press CTRL+C to quit
Invoking app.lambda_handler (python3.11)
Reuse the created warm container for Lambda function 'HelloWorldFunction'
Lambda function 'HelloWorldFunction' is already running
START RequestId: 0652f01c-89a7-43d9-85fb-3844c8f1a32b Version: $LATEST
END RequestId: b229bc70-c5b6-4bf2-b9a3-97b2cf9340a1
REPORT RequestId: b229bc70-c5b6-4bf2-b9a3-97b2cf9340a1 Init Duration: 0.04 ms Duration: 1153.99 ms Billed Duration: 1154 ms Memory
Size: 128 MB Max Memory Used: 128 MB

No Content-Type given. Defaulting to 'application/json'.
2024-04-03 09:26:38 127.0.0.1 - - [03/Apr/2024 09:26:38] "GET /hello HTTP/1.1" 200 -
2024-04-03 09:26:39 127.0.0.1 - - [03/Apr/2024 09:26:39] "GET /favicon.ico HTTP/1.1" 403 -
```

Output:



Practical 8

AIM: Implementation of Openstack with user and private network creation.

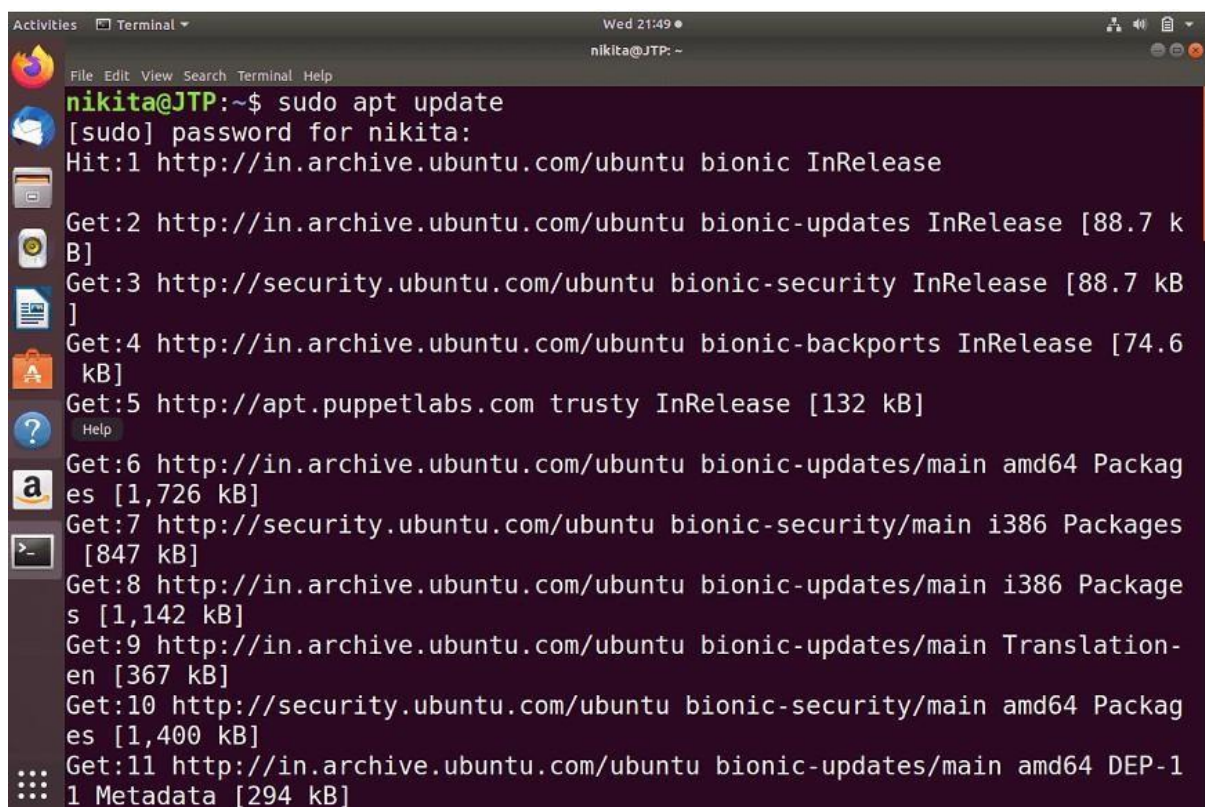
STEPS:

Step 1: Update Ubuntu System

Open the terminal and run the following command to ensure that the system is up to date :

- \$ sudo apt update **OR**
- \$ sudo apt -y upgrade

Sample Output :



```
nikita@JTP:~$ sudo apt update
[sudo] password for nikita:
Hit:1 http://in.archive.ubuntu.com/ubuntu bionic InRelease
Get:2 http://in.archive.ubuntu.com/ubuntu bionic-updates InRelease [88.7 k
B]
Get:3 http://security.ubuntu.com/ubuntu bionic-security InRelease [88.7 kB
]
Get:4 http://in.archive.ubuntu.com/ubuntu bionic-backports InRelease [74.6
kB]
Get:5 http://apt.puppetlabs.com trusty InRelease [132 kB]
Get:6 http://in.archive.ubuntu.com/ubuntu bionic-updates/main amd64 Packag
es [1,726 kB]
Get:7 http://security.ubuntu.com/ubuntu bionic-security/main i386 Packages
[847 kB]
Get:8 http://in.archive.ubuntu.com/ubuntu bionic-updates/main i386 Package
s [1,142 kB]
Get:9 http://in.archive.ubuntu.com/ubuntu bionic-updates/main Translation-
en [367 kB]
Get:10 http://security.ubuntu.com/ubuntu bionic-security/main amd64 Packag
es [1,400 kB]
Get:11 http://in.archive.ubuntu.com/ubuntu bionic-updates/main amd64 DEP-1
1 Metadata [294 kB]
```

Reboot the system after running the above command. To reboot the system, run the following command :

- \$ sudo reboot

or

- \$ init 6

Step 2: Create Stack User

It is important that the devstack must run as a regular user (non-root user) with the sudo enabled.

To keep this note in mind, let's create a new user with the name "stack" and assign the sudo permissions or privileges. To create a stack user, run the following command in your terminal:

- `$ sudo useradd -s /bin/bash -d /opt/stack -m stack`

Output :

```
nikita@JTP:~$ sudo useradd -s /bin/bash -d /opt/stack -m stack
nikita@JTP:~$
```

Now, to assign the sudo privileges to the stack user, run the following command :

- `$ echo "stack ALL=(ALL) NOPASSWD: ALL" | sudo tee /etc/sudoers.d/stack`

Output :

```
nikita@JTP:~$ echo "stack ALL=(ALL) NOPASSWD:ALL" | sudo tee /etc/sudoers.d/stack
stack ALL=(ALL) NOPASSWD:ALL
nikita@JTP:~$
```

You can switch to the 'stack' user by running the following command:

- `$ sudo su - stack`

Output :

```
nikita@JTP:~$ sudo su - stack
stack@JTP:~$
```

```
nmfc@nmfc-virtual-machine:~/Desktop$ echo "stack1 AL=(ALL) NOPASSWD:ALL" | sudo
tee /etc/sudoers.d/stack
[sudo] password for nmfc:
stack1 AL=(ALL) NOPASSWD:ALL
nmfc@nmfc-virtual-machine:~/Desktop$ sudo apt install git -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  git-man liberror-perl
Suggested packages:
  git-daemon-run | git-daemon-sysvinit git-doc git-email git-gui gitk gitweb
  git-cvs git-mediawiki git-svn
The following NEW packages will be installed:
  git git-man liberror-perl
0 upgraded, 3 newly installed, 0 to remove and 563 not upgraded.
```

Step 3: Install the Git

In Most of the Ubuntu systems, git comes by default. But if git is missing on your system, then install it by running the following command:

- `$ sudo apt install git -y`

Sample Output :

```

stack@JTP:~$ sudo apt install git -y
Reading package lists... Done
Building dependency tree
Reading state information... Done
The following packages were automatically installed and are no longer required:
  libllvm9 linux-headers-5.3.0-28 linux-headers-5.3.0-28-generic linux-image-5.3.0-28-generic
  linux-modules-5.3.0-28-generic linux-modules-extra-5.3.0-28-generic
Use 'sudo apt autoremove' to remove them.
The following additional packages will be installed:
  git-man liberror-perl
Suggested packages:
  git-daemon-run | git-daemon-sysvinit git-doc git-el git-email git-gui gitk gitweb git-cvs git-mediawiki
  git-svn
The following NEW packages will be installed:
  git git-man liberror-perl
0 upgraded, 3 newly installed, 0 to remove and 0 not upgraded.
Need to get 4,741 kB of archives.
After this operation, 34.0 MB of additional disk space will be used.
Get:1 http://in.archive.ubuntu.com/ubuntu bionic/main amd64 liberror-perl all 0.17025-1 [22.8 kB]
Get:2 http://in.archive.ubuntu.com/ubuntu bionic-updates/main amd64 git-man all 1:2.17.1-1ubuntu0.7 [804 kB]
Get:3 http://in.archive.ubuntu.com/ubuntu bionic-updates/main amd64 git amd64 1:2.17.1-1ubuntu0.7 [3,915 kB]
Fetched 4,741 kB in 58s (81.9 kB/s)
Selecting previously unselected package liberror-perl.
(Reading database ... 205033 files and directories currently installed.)
Preparing to unpack .../liberror-perl_0.17025-1_all.deb ...
Unpacking liberror-perl (0.17025-1) ...
  
```

Step 4: Download OpenStack

Once you install the git, use the git command to download the DevStack from Github.

```
$cd devstack
```

```
$cp samples/local.conf .
```

- ```
$ git clone https://git.openstack.org/openstack-dev/devstack
```

Output :

```

stack@nmfc-virtual-machine:~$ cd devstack
cp samples/local.conf .
stack@nmfc-virtual-machine:~/devstack$ git clone https://git.openstack.org/openstack-dev/devstack
Cloning into 'devstack'...
warning: redirecting to https://opendev.org/openstack/devstack/
remote: Enumerating objects: 52323, done.
remote: Counting objects: 100% (31913/31913), done.
remote: Compressing objects: 100% (10895/10895), done.
remote: Total 52323 (delta 31135), reused 21018 (delta 21018), pack-reused 20410
Receiving objects: 100% (52323/52323), 9.81 MiB | 2.77 MiB/s, done.
Resolving deltas: 100% (37149/37149), done.
stack@nmfc-virtual-machine:~/devstack$

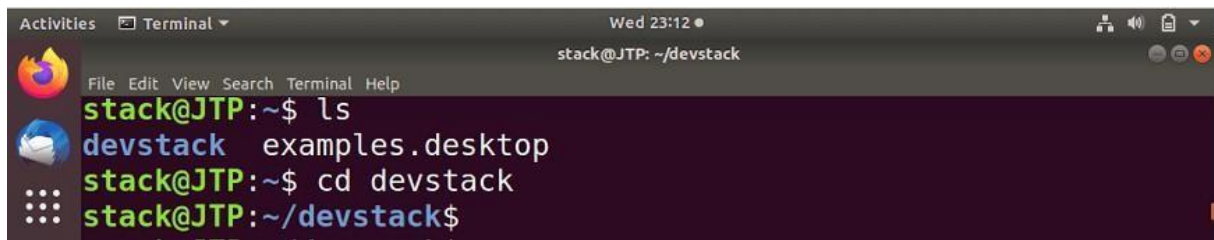
```

#### Step 5: Create a DevStack Configuration File

First of all, go to the devstack directory by running the following command :

- ```
$ cd devstack
```

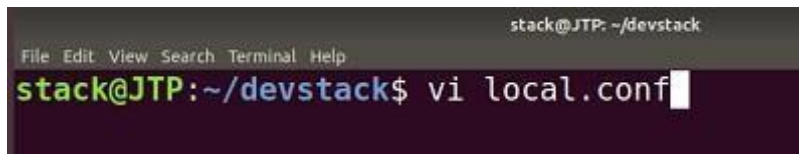
Output :

A terminal window titled 'stack@JTP: ~/devstack' showing the following commands and output:

```
stack@JTP:~$ ls
devstack  examples.desktop
stack@JTP:~$ cd devstack
stack@JTP:~/devstack$
```

Now, create a local.conf file in which you have to enter the four passwords and the host IP address :

Output :

A terminal window titled 'stack@JTP: ~/devstack' showing the command:

```
stack@JTP:~/devstack$ vi local.conf
```

Copy the following line of content in the file :

- `[[local|localrc]]`
-
- `# Password for KeyStone, Database, RabbitMQ and Service`
- `ADMIN_PASSWORD=StrongAdminSecret`
- `DATABASE_PASSWORD=$ADMIN_PASSWORD`
- `RABBIT_PASSWORD=$ADMIN_PASSWORD`
- `SERVICE_PASSWORD=$ADMIN_PASSWORD`
-
- `# Host IP - To get your Server or VM IP, run the 'ip addr' or 'ifconfig' command`
- `HOST_IP=192.168.56.103`

Output :

A terminal window titled 'stack@JTP: ~/devstack' showing the content of the local.conf file:

```
[[local|localrc]]

#Password for KeyStone, Database, RabbitMQ and Service
ADMIN_PASSWORD=StrongAdminSecret
DATABASE_PASSWORD=$ADMIN_PASSWORD
RABBIT_PASSWORD=$ADMIN_PASSWORD
SERVICE_PASSWORD=$ADMIN_PASSWORD

#Host IP - To get your server or VM IP run the 'ip addr' or 'ifconfig' command
HOST_IP=192.168.56.103
~
~
~
```

Press the ESC, then :wq and then press enter to save and then exit from the local.conf file.

Here, ADMIN_PASSWORD is the password that we will use to log into the OpenStack login page. The default username for an OpenStack is 'admin'.

And HOST_IP is the IP address of your system. To get your Server or VM IP, run the 'ifconfig' or 'ip addr' command.

Step 6 : Install OpenStack with DevStack

To install and run the openstack, execute the following command :

- \$./stack.sh

DevStack will install the following components:

- Compute Service (Nova)
- Image Service- Glance
- Identity Service-Keystone,
- Block Storage Service - Cinder
- OpenStack Dashboard - Horizon
- Network Service - Neutron
- Placement API - Placement
- Object Storage - Swift

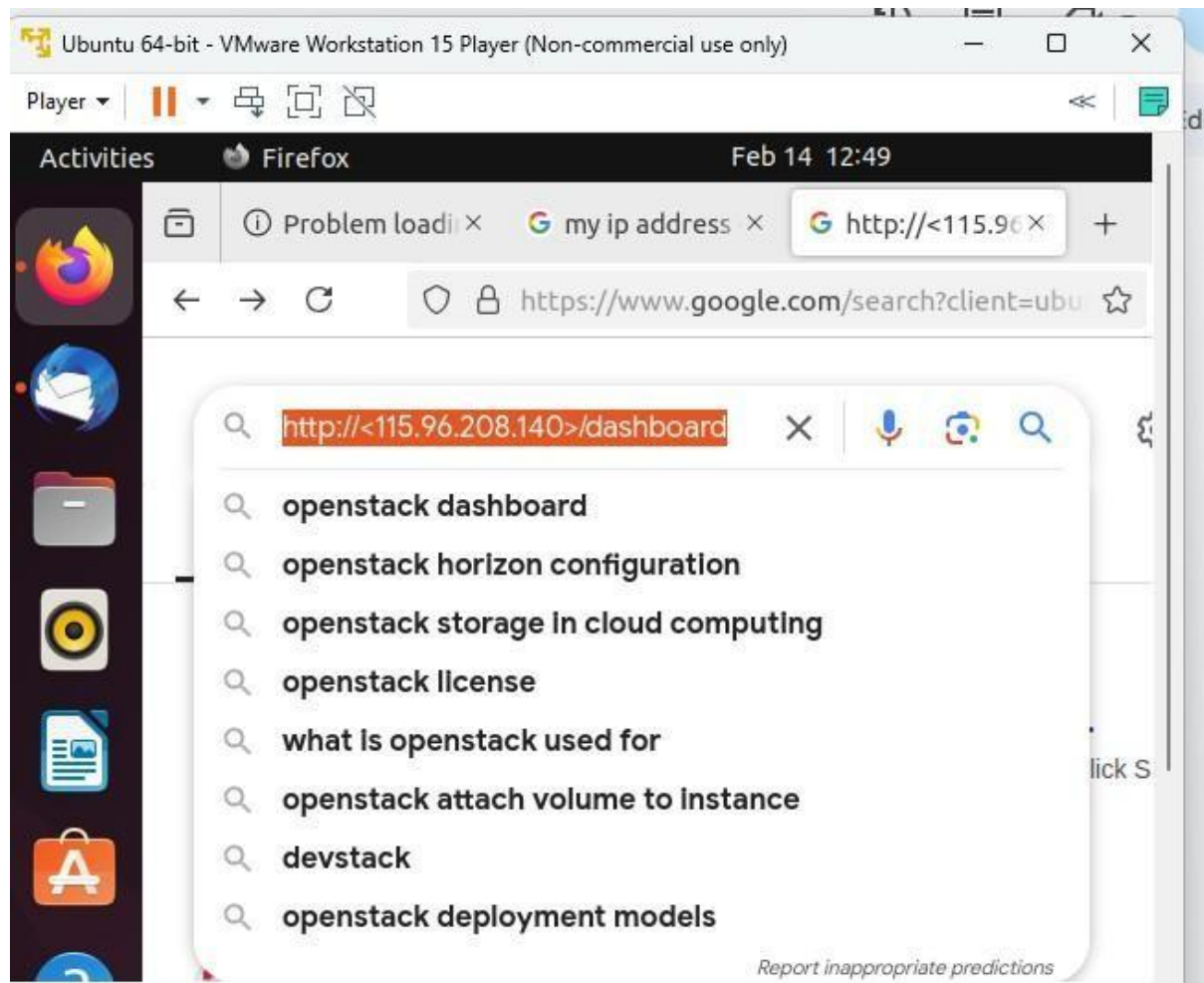
The installation will take about 10-20 minutes, mostly depends on your internet speed.

At the very end of the installation, you will get the host's IP address, URL for managing it and the username and password to handle the administrative task.

Step 7: Accessing OpenStack on a browser

Copy the horizon URL given in the installation output and paste it into your browser :

- http://IP Address/dashboard



Conclusion: Hence we have successfully implemented Openstack with user and private network creation.